



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

NATURAL SCIENCES YEAR 2 TERM 1 TO 4 LEARNER'S GUIDE



LIFE AND LIVING

CONTENT OVERVIEW

LIFE AND LIVING	<u>Living and non-living things</u> Living things • <u>Identify</u> and explain • Briefly explain the seven life processes • Non-living • Identify • Explain why things are non-living Practical: Living and Non-living things	<u>Digestive system</u> • Explain concepts • Describe structure • Name the parts and the organs • Describe the features and functions of the parts and organs	<u>Sense Organs</u> Senses • Briefly define concepts • Name and describe animal senses adapted for survival
	<u>Structure of plants</u> • Describe the basic structure of plants • Identify differences between plants	<u>Nutrients in food</u> Food Groups • Briefly define concepts • Categorise food • Describe the functions of food groups	<u>Human skeleton</u> • Briefly define concepts • Function of bones
	<u>Structure of animals</u> • Describe the basic structure of animals • Identify differences between animals	<u>Nutrition</u> • Explain a balanced diet	<u>Human Reproductive System</u> Reproduction • Introduce and explain concepts with examples • Explain the physical changes • Explain the features in the human organisms
	<u>What plants need to grow</u> • Explain requirements to growth • Explain how plants grow from cuttings and seeds	<u>Food processing</u> • Explain concepts, importance of preserving, types of products, methods and benefits	

CONTENT OUTLINE

WEEK	TOPIC	CONTENT
1-3	Living and Non-living things that share the world with us	Living things - Identify and explain different kinds of living things - Explain all seven life processes: feeding, growing, reproducing, breathing, excreting, sensing, and moving carried out by living plants and animals - Explain with examples why some things appear to be dead but will come alive, given the right conditions, e.g. dried yeast, and dried beans, an incubated fertilized egg. Explain why some things were living and are now dead: e.g. dead wood/leaves. Non-living things - Identify and explain different kinds of non-living things: - Explain why some things were living and are now dead: e.g. dead wood, dry leaves. Practical activity: Identify and explain the seven life processes of any living organism.
4-5	Structure of plants	Features of plants - Describe the following concepts with examples: roots, stems, leaves, flowers, fruits, seeds. - Describe the basic structure of plants: roots, stems, leaves, flowers, fruits, seeds - Identify visible differences and similarities between plants: e.g. small, large, grow in water or soil, produce fruits, flowers; thick/thin/large/small leaves etc. Practical demonstration: Learners must touch, smell and observe the differences in plant structures.
6	What plants need to grow	Requirements for growth of plants - Explain the requirements for plants to grow: light, warmth, water, gases and fertile soil. - Explain and demonstrate how new plants can be grown from cuttings or seeds. Practical Investigation: Growth of plants from seeds and

		cuttings.
7-8	Structure of animals	Structure of Animals • Explain the following concepts with examples: Head, Tail, Body, Limbs, Sense organs, Body covering, Vertebrate, Invertebrate Features of animals • Describe the basic structure of animals: head, tail, body, limbs, sense organs, body covering, and vertebrate, invertebrate. • Identify visible differences between vertebrate and invertebrate animals.
9 - 10	The weeks allocated for formal assessment are integrated across the term.	
Formal Assessment: • Practical: 50 % Test (Pen and Paper): 50 %		

TOPIC 1: LIVING AND NON LIVING THINGS

KEY QUESTIONS

- What does it mean to be alive?
- What is a non-living thing?
- What does it mean to be non-living?
- A chicken egg seems to be non-living, but then it can hatch into a chicken. Is the egg living or non-living?

1.1. LIVING THINGS

There is a wide variety of livings, including plants and animals. They are distinguished from non-living things by their ability to move, reproduce and are sensitive towards their environment. To survive they need water, food, a way of getting rid of waste and an environment which stays within a particular range of temperature.

INSTRUCTIONS

1. Look through these photos and decide whether you think they are living or non-living.
2. When you are done, discuss your choices with your class.



Flowers and plants



Water and waves ¹



A zebra



Clouds in the sky ²



There are things that look as if they are non-living, which are living given the right conditions for example seeds, yeast and mould.

Characteristics of living things

Living things may look different, but they all carry out seven life processes.

Let's take a look at each of these.

MOVING: All living plants and animals move

- Humans and animals use their bodies to move from one place to another.
- Some plants turn towards light or water. Roots mostly grow downwards. Many stems grow upwards.



Humans move all the time. These athletes are running. 7

REPRODUCTION: All living things reproduce (babies or seeds)

- Humans and animals have babies.
- Some new plants can grow from seeds.
- Other plants grow from cuttings or shoots.



A mother and father with their baby

SENSING: All living things respond to any change that they sense

- When you are feeling cold, you will put on a jersey or jacket.
- When it becomes winter some animals hibernate.
- In autumn the leaves on some trees change colour.
- You can use an umbrella to protect you from the rain or from the harsh sun on a hot day.
- Reptiles like to lie and bask in the sun on cold winter's days.



This chameleon is basking on the wall in the winter sun.



The leaves on some trees respond to the change of the season. These leaves are turning brown during autumn.

RESPIRATION: All living things exchange gases in and out

- Humans and animals use the gas, oxygen from the air that they **breathe** in.
- They release (give off) the gas carbon dioxide when they breathe out.
- During respiration plants take in the gas carbon dioxide into their leaves. They use it to make food. They then release oxygen for animals and humans to use.

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FEEDING: All living things need food

- Food gives all living things the energy they need.
- Green plants can make their own food for energy in their leaves and stems.



Picture retrieved from: [\(1293\) Pinterest](#)

EXCRETING: All living animals and plants have to get rid of waste products

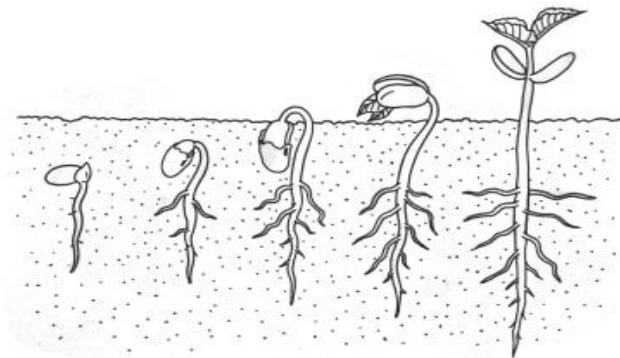
- Humans and animals have to get rid of waste products from their bodies.
- Plants get rid of waste water through the process of transpiration.



Do you see how shiny the horse looks? She is sweating from all that running! ⁹

GROWING: All living things grow

- Human and animal babies grow into adults.
- Seedlings grow into plants.



Growth of a seed into a seedling

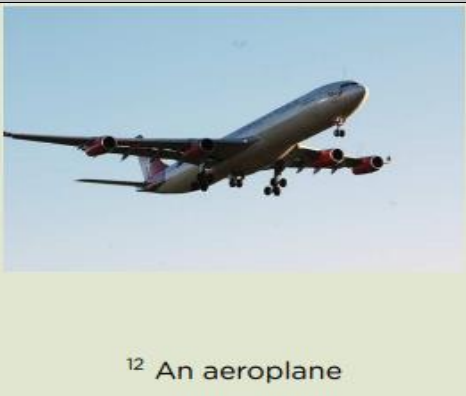
All seven of the life processes must happen for something **to be living**. If something does not carry out all seven life processes, then it is non-living.

ACTIVITY 1: Understanding the seven life processes

THE REASON FOR DOING THIS ACTIVITY: To help you understand the seven life processes.

INSTRUCTIONS:

1. Look carefully at each photo.
2. Decide whether the object is living or non-living and write your answer in the middle column.
3. Provide a reason for your answer in the last column.

Object	Living or non-living?	Reason
 12 An aeroplane		



¹³ Fish in the sea



¹⁴ A plant



¹⁵ A bouncing soccer ball



¹⁶ Chicken eggs

Some things seem to be non-living, but they are not!

Yes, some things seem to be non-living for a very long time. When exposed to the right conditions they revive again, they show the characteristics of living things.

Look at the picture below of seeds which seem to be non-living!



Picture retrieved from: [Dry Bean Seed - AGT Foods](#)

QUESTIONS

1. Why do seeds seem to be non-living? How can we show that they are living?

We say the seeds are in a "dormant state" until they are given water, warmth, air and light to germinate and start growing. There are other things too which seem to be non-living, but if they are given the right conditions then they can revive and carry on living.



A dove keeping her eggs warm to hatch them.

Fertilised eggs need to be kept warm or they will not hatch. This is why a mother bird will start sitting on her eggs to keep them warm when she has laid all of them.



Retrieved from: [How Yeast Is Used in Cooking \(thespruceeats.com\)](http://thespruceeats.com) ; [What every new baker should know about the yeast all around us \(theconversation.com\)](http://theconversation.com)

Yeast can cause bread dough or cake batter to raise.

ACTIVITY: Can I revive living things that seem to be non-living?

MATERIALS (what you need):
Photos of eggs hatching

INSTRUCTIONS:

1. Look at these photos carefully.



Chicken eggs hatching. ¹⁹



Frog eggs about to hatch into tadpoles. ²⁰



Shark eggs. ²¹



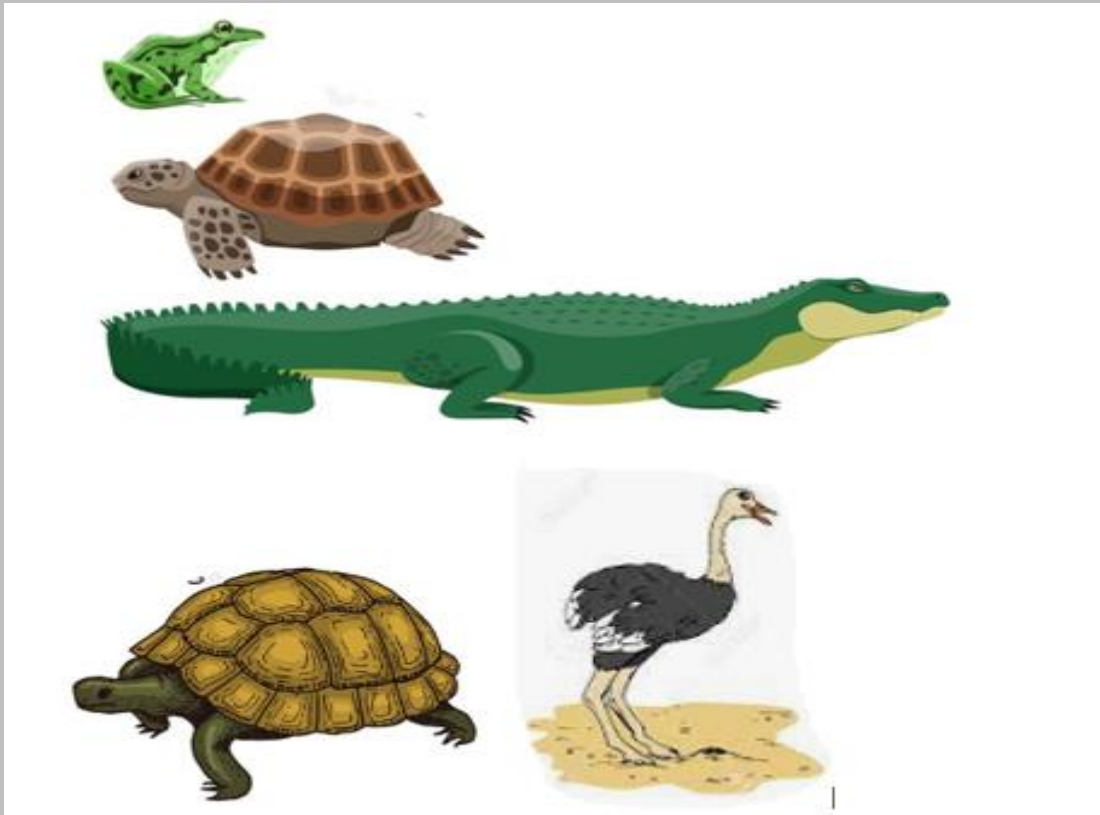
Snake eggs hatching. ²²

QUESTIONS:

1. Study the photo of the bird sitting on her nest. Can you explain why she needs to sit on her eggs?
2. Farmers often do not let the mother chickens sit on their eggs. Instead they put the eggs in something called an incubator, as you see in the picture above of the chicken eggs hatching. What does the incubator provide to the eggs?
3. A snake normally lays her eggs in a 'nest'. Why does she not have to lie on top of them to hatch them?
4. Have you ever caught tadpoles or kept silkworms in a box?

5. RESEARCH QUESTION:

South Africa has a wide variety of animal laying eggs.



How do different animals hatch their eggs?

Frog turtle/tortoise crocodile ostrich

Your answer need to include the following information

1. Where do they lay the eggs?
2. When do they lay the eggs?
3. How do they hatch?
4. How long does it take for the eggs to hatch?
5. Provide pictures.

In your class group discuss all the similarities and differences between the ways in which the animals hatch their eggs.

Animal	Research Information

ACTIVITY: Germinating a seed (To show that a seed is a living thing)**MATERIALS:** (what you need):

- bean seeds
- cotton wool
- plastic lids (from empty peanut butter jars for example)

INSTRUCTIONS:

1. Place two layers of cotton wool in the plastic lid.
2. Place a few of your seeds between the two layers of cotton
3. Drizzle water over the seeds. You need to water the cotton wool enough to wet it but NOT TO DROWN THE SEEDS! There should be NO WATER running over the sides of the lid or your seed will drown!
4. Place your seeds in a warm place near a window.
5. Water your seeds whenever you feel the cotton wool is almost dry.

Be careful not to drown your seeds!

QUESTIONS:

1. Before you water your seeds, describe how they look and feel.
2. Draw your seeds between the cotton wool on the first day.
3. Check the progress of your seeds every day. How long did it take them to germinate? And what do your seeds look and feel like now?
4. What do you think made your seeds revive?

ACTIVITY: DEMONSTRATION (Getting yeast to grow!)**MATERIALS** (what you need):

- packet of dry yeast
- sugar
- warm water
- an empty yogurt tub/ any transparent container

INSTRUCTIONS:

1. Place a teaspoon of sugar and a teaspoon of dry yeast in your yogurt tub. Mix with your spoon.
2. Add 3 teaspoons of warm water.
3. Stir your sugar and yeast mixture in the warm water to make sure it is well mixed.
4. Watch to see what will happen!

SAFETY WARNING! Don't use boiling water - it might burn you! Boiling water will also kill the yeast. You only need to use lukewarm water!

QUESTIONS:

1. What does your yeast look and feel like before you mix it with the sugar and water?
2. When you add the sugar to the yeast, does anything change in the yeast?
3. What happened to the yeast and sugar mixture when you added the warm water?
4. How did the yeast revive?

1.2. NON-LIVING THINGS

Non-living things are different from living things because they do not perform seven life processes.

ACTIVITY: Distinguishing between living and non-living things

The reason for doing this activity:

- To help you learn to distinguish between living and non-living things.

MATERIALS (what you need):

- 3 - 5 different pictures with objects that are living or non-living (bring them to school)
- Scrap paper or cardboard

INSTRUCTIONS:

1. Use the scrap paper or cardboard to make four labels of the following headings:
 - Living
 - Non-living
2. Show the pictures or objects you brought to your group. Place each item or picture under one of the headings you just made.
3. Now complete the table below with the results. If there is time left over you can add in interesting objects from other groups into your table as well.

Living	Once lived	Can be revived	Never lived
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4. Carefully look at these photos. Say which of these are living or non-living, or which was once living or can be revived. Write these labels below each of these.



A dog



Traffic lights



Eggs



Burning fire



Trees



Paper



A dolphin



A computer



A skull

KEY CONCEPTS

- We can group things on Earth as living and non-living.
- There are seven life processes that all living things carry out.
- Non-living things cannot carry out seven life processes.
- Living things can die.
- Some things like seeds or eggs can seem to be non-living but they are living

REVISION



Picture retrieved from: [What living and non living things can you spot in this picture? | Kingfishers' Class Blog \(wordpress.com\)](http://www.kingfishersclassblog.wordpress.com)

QUESTIONS

1. Name five non-living things you can identify in the picture.
2. Name five things in the picture that use oxygen.
3. What life process in living things uses oxygen?
4. Give an example from the picture of:
 - a. moving:
 - b. sensing:
 - c. feeding:
 - d. growth:
5. What life process do we associate with the eggs in the nest?

TOPIC 2: STRUCTURE OF PLANTS:

KEY QUESTIONS

- Are all plants made in the same way with the same parts?
- If I cannot see leaves on a cactus, is it still a plant?
- Is the moss that grows near an outside tap a plant? How can I tell if it is?
- If a cactus, seaweed and a dandelion are all so different, how can you say they are all plants?

We learnt that almost everything can be grouped into two groups - living and non-living. In the next section we are going to learn more about living plants and animals, how they look and what makes them special!

BASIC STRUCTURES OF PLANTS

All plants have the following basic parts:

- roots
- stems
- leaves

Let's take a look at the different (parts of the plant

Roots

Plant roots are normally found underground. Roots have very important functions:

- roots anchor the plant in the ground
- roots absorb water and nutrients from the soil, which are then transported to the rest of the plant
- some plants store the food they make in their roots, like carrots. Later on you will see how plants make their own food!



The roots of this tree go deep down into the soil.



A carrot is a root that stores food made by the plant

Picture of tree: [Are Tree Roots Messing with Your Home's Mojo? - King Piers LLC](#)

Stems

Stems connect the roots to the rest of the plant. The stem has important functions:

- the stem supports the leaves, flowers and fruit
- the stem carries nutrients and water from the roots to the other parts of the plant
- some plants store the food they produce in their stems (like sugar cane or asparagus or potato)



A growing plant stem.¹



A tomato plant stem.

Leaves

Although many plants' leaves are green, leaves can have many other colours. Some leaves change colour during autumn. Leaves have the following functions:

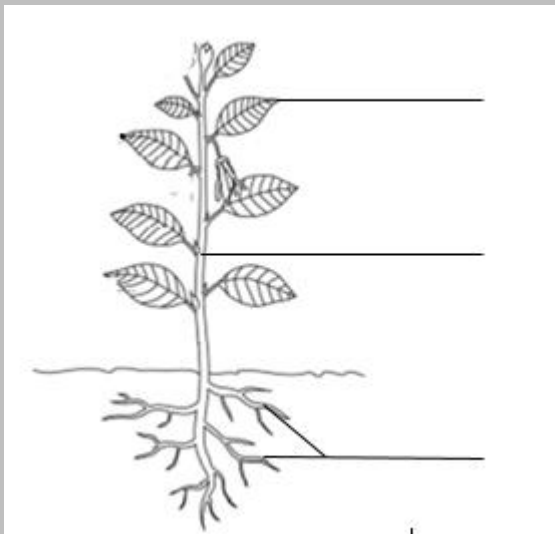
- leaves absorb sunlight and use it to make food for the plant.
- some leaves store water.
- most leaves have veins which are like tiny pipes that carry water and nutrients from the roots (the veins carry the food the leaf makes to the rest of the plant)



Can you see the veins in these leaves? ²

Stem, roots and leaves are the basic parts of all plants. Plants **can** have flowers, fruits and seeds.

Label the structure of the plant below



Visible differences between plants

There are many different kinds of plants. If you look at different plants you can see many things that are different but also things that are the same. We know that most plants have stems, roots and leaves, and that others have flowers, seeds and fruit. If we want to compare plants, we can compare these plant structures. You can look at the different parts of plants and compare their:

- size
- colour
- shape

Or you can ask really important questions about the plants, like:

- Does this plant flower?

- Does it lose its leaves in autumn?
- Can animals eat the plant or parts of the plant?
- Can humans eat the plant?
- Perhaps you can think of other important questions that you could ask

We are going to compare different plants using our eyes as our guides.

Look at the photos of the banana palm and the basil plant. How many differences can you see between these two plants?



A banana palm³



A basil plant.

When we compare plants, it is sometimes easier to use the different plant structures to compare the plants. We can look at the stems, for example, in the banana palm and the basil plant and compare them. The basil plant has a thin green stem while the banana palm has a thick brown woody bark covering its very thick trunk.

Look at the photograph below of the edge of a river. There are main types of plants growing: the water lilies in the front and the reeds at the back. Both grow near or in water but they look completely different!

ACTIVITY: Comparing plants according to their structures



Waterlilies and reeds growing in water.



Tambo / Phragmites vulgaris / COMMON REED. Lu gen: Philippine Medicinal Herbs / Philippine Alternative Medicine (stuartxchange.com)



water lily roots stem leaves - Google Search

INSTRUCTIONS:

1. Work with a friend.
2. Compare the two plants (water lilies and reeds) using the plant structures.
3. When people compare different things using a set of items (like the plant structures we are using), they often use a table to write down their ideas.
4. Study the table below. Write differences and similarities between the plant structures of the water lilies and the reeds.

	Water lilies	Reeds
Stems		
Roots		
Leaves		

TOPIC 3: WHAT PLANTS NEED TO GROW?

Conditions for growth

- How can I grow my own plants?
- If I plant seeds, what must I do to make sure that they grow?
- What does a plant actually need to stay alive and grow?

What do plants need to grow?

In this section we are going to learn specifically what makes plants grow. We will also look at growing new plants and how you can make sure that as many of your seeds as possible grow into healthy plants.

QUESTIONS

After the activity when you planted a bean seed, how did your bean grow - did it die or did it stay alive? Discuss what you think your bean plant will need to stay alive and continue growing.

1. Sunlight

- Plants need sunlight to grow
- Green plants use sunlight, water and carbon dioxide gas to make food.
- The plant can use some of this food to grow

2. Air

- Plants need air to grow. Plants use carbon dioxide in the air to make food so that they can grow.

3. Water

- Plants need water to grow. Some plants need more water than others. The amount of water a plant needs depends on the type of plant. If the plant does not get the amount of water it needs it will die. Some plants are able to grow in very dry areas, such as cacti in the desert. These plants have adapted (changed) over many, many years to be able to survive in these conditions.
- The roots of the plant absorb water from the soil. This water carries nutrients from the soil to all the parts of the plant.

Growing new plants from cuttings

We already dealt with the planting of the bean seed in the first topic.

Plants can generally be grown from seeds or cuttings.

- A cutting is made when a piece of a plant (usually the stem) is cut off and planted in new soil to start growing roots and form a new plant.

QUESTIONS

1. What three really important things do plants need to grow?
2. Do you remember that one of the life processes is reproduction? How do plants make new plants?

TOPIC 4: STRUCTURE OF ANIMALS

In this section we are going to learn how to identify different animals. Then we will compare some animals you might already know.

Basic structures in animals

Let's take a closer look at the body parts of animals. Just like plants, animals also have a basic structure. The basic structure of an animal is:

- head
- tail
- body
- limbs
- sense organs

Head

Most animals have a part of their body that we call the 'head'. Even the smallest animal has a part where it's 'brain' is. In most animals the head has:

- a brain (or brain-like structure) - no matter how small
- Sensory organs (like the eyes and ears)
- feeding parts (like the mouth and jaws)

Tail

Most animals have a tail at the back end of their body (have you ever wondered where a starfish or octopus' tail is?)

- A tail is often pointed but can have many other shapes as well.

Body

Different animals have different body structures and body covering.

Limbs

Most animals use their limbs to move. Animals can walk, run, climb or swim using their limbs. Some animals like mice and squirrels can use their front or upper limbs to handle objects.

Animals can have wings, webbed feet, tentacles, fins, legs, arms, flippers and long slithery bodies.

Senses

Animals have senses that help them to sense the environment around them.

Sniffer dogs help to find people who are trapped under building rubble, mudslides or snow and tell the rescue workers where the victims are. These dogs also smell drugs or bombs and alert the police.



Sniffer dogs at the airport. ¹⁵

- Eagles, buzzards, hawks and other birds of prey have extremely sharp eyes as they have to see small rodents from very far away.



[Little Owl - eBird](#)

- Elephants, cats and dogs can hear sounds that human ears cannot hear.



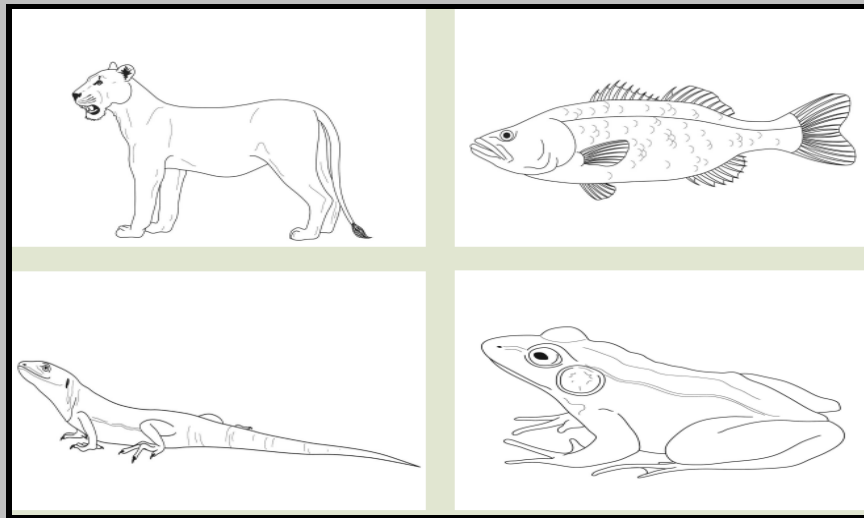
ACTIVITY: Body plans of animals

THIS REASON FOR DOING THIS ACTIVITY:

When you were learning about plants, you also learnt how to label a drawing in a scientific way. In this activity you are going to practice your labelling skills.

INSTRUCTIONS:

1. Study each of these drawings of different animals.
2. Use your scientific labelling skills to label each animal using the five body structures of animals



Grouping animals

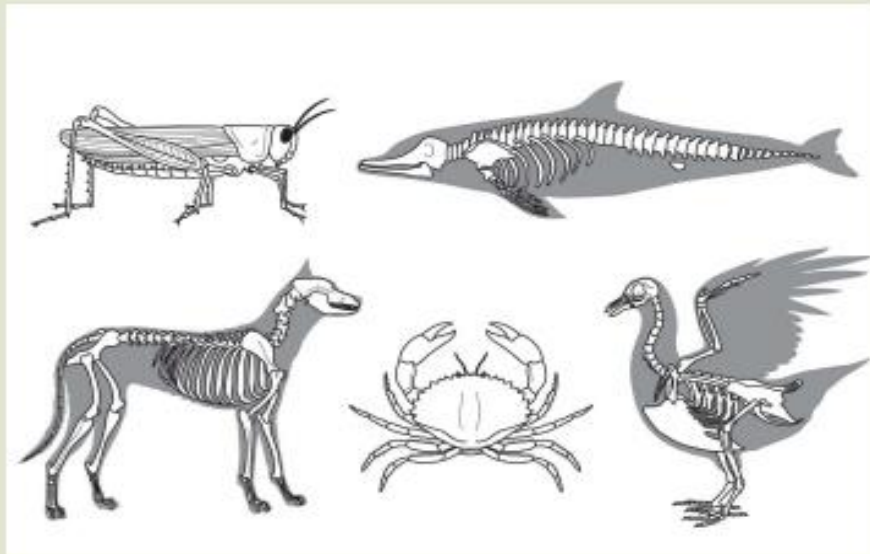
When we group similar things together, it is called classifying. When classifying animals, there are generally two main groups of animals - those who have bones inside their bodies with a backbone, and those who do not have bones inside their bodies.

- Animals **with a backbone** are classified as **vertebrates**.
- Animals **without a backbone** are classified as **invertebrate**.

ACTIVITY: Classifying animals

INSTRUCTIONS:

1. Carefully look at the following pictures and decide which animals are vertebrates and which are invertebrates.



Examples of vertebrates and invertebrates

Animals that are vertebrates	Animals that are invertebrates:

2. Identify the visible differences between the vertebrates and invertebrates listed above.

Vertebrates	Invertebrates

TERM 2: MATTER AND MATERIALS

CONTENT OVERVIEW TERM

MATTER AND MATERIALS	<u>States of matter</u> Solids, liquids and gases - Briefly define and describe three states - Distinguish between states - Describe the properties	<u>Mixtures</u> - Briefly define concepts - Distinguish between pure substances and mixtures - Separating of mixtures (demonstrate)	<u>Acids and bases</u> - Briefly define concepts - Describe difference between acids and bases - Test various household substances with blue and red litmus paper and sort them into acids, bases and neutrals Practical activity
	<u>Metals and non-metals</u> - Define concepts - Distinguish between metals and non-metals - Properties of metals and non-metals - Uses of metals and non-metals	<u>Solutions as special mixtures</u> - Ingredients of a mixture - Explain the difference between a mixture and solution - Describe the properties of a mixture	<u>Process to purify water</u> Clean Water - Introduce and explain concepts with examples - Importance of drinking clean water - Methods of cleaning water Practical Investigation
		<u>Dissolving</u> - Explain concept - Rate of dissolving Practical investigation: Rate of dissolving	<u>Impact of weather, temperature and wind on matter and materials.</u> - Impact of weather on materials - Ways to protect materials
	<u>Processing material</u> Briefly define	<u>The water cycle</u> Briefly define concepts	

	concepts • Explain processes to make new materials or products • List the properties of raw materials used to make new products Practical: Demonstrate the changes of raw material	• Describe the stages of the water cycle • Draw the water cycle and describe in your own words Practical	
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CONTENT OUTLINE: TERM 2

WEEK	TOPIC	CONTENT
1-3	States of matter	Solids, Liquids and gases - Briefly define concepts with examples: Solid, Liquid, Gas, Evaporation, Condensation, Freezing, Melting - Demonstrate practically the 3 states in which matter could be found (solids, liquids, gases) - Distinguish between solids, liquids and gases with practical examples. - Describe the properties of solids, liquids and gases. - Describe and explore how matter can change into different states through heating and cooling using different applicable materials.
4-6	Metals and non-metals	Properties of metals - Briefly define concepts: Metals; Polymers; Ceramics - Distinguish between metals and non-metals - Name different materials in the immediate surroundings and describe their features (e.g. colour, texture etc.) - Describe the properties of different classes of metals: Appearance; Flexibility; Hardness; Shiny; Strength; Heat conduction - Describe the properties of non-metals: Dullness; Can break easy; Heat conduction; Ability to rust Uses of metals and non-metals - Research and describe how metals and non-metals are used to make products for example furniture made in the workshops at school.
7-8	Processing material	Briefly explain concepts: - Combining, Mixing, Setting, Cooking/Heating, Cooling, Drying - Explain that all materials can be processed to make new materials or products. - Explain that in Africa people have processed materials e.g. clay pots and bricks etc. - Explain that processed materials are useful because of their special properties: strong, durable, waterproof, different colours etc. - List different products and list the different raw materials it is made of. Practical: Activity - Demonstrate and explain processes used to change the properties of raw materials:

		- Mixing and setting - Mixing - Mixing and cooking - Mixing and cooling
9-10	The weeks allocated for formal assessment are integrated across the term.	
Formal Assessment: • Practical: 40 % • Test (Pen and Paper): 60 %		

TOPIC 1: STATES OF MATTER

1.1 SOLIDS, LIQUIDS AND GASES

KEY QUESTIONS

- What are solids, liquids and gases?
- What is change of state?

Everything around us is made up of matter. Solids, liquids and gases in the universe are matter. Matter takes up space and has mass, this means we can weigh matter. When we use one kind of matter to make something such as a wooden or plastic chair we say the material used was wood or plastic.

DEFINITION OF CONCEPTS

Solid

Substance that is firm and stable in shape. Solids keep their shape. Solids could be hard, soft, big or small. Some examples are stone, wood, towel and sand.

Liquid

Substance that flows freely but is of constant volume. Liquids flow and take on the shape of the container they are in. Some examples are water, paraffin, baby oil, fruit juice, petrol or methylated spirits

Gas

Substance that expands to fill the space in it. No fixed shape and no fixed volume. Some examples are helium, oxygen and carbon dioxide

Evaporation

The process of changing from liquid into gas.

Evaporation takes place when heat is added to the liquid.

Condensation

Process of turning from gas to liquid. It is the reverse of evaporation.

Condensation takes place when heat is removed.

Freezing / Solidifying

When a liquid change to a solid.

Melting

When a solid change to a liquid.

Practical demonstration of the 3 states in which matter can be found

ACTIVITY

Exploring the properties of solids

Materials:

- a stone
- cloth
- paper
- a table or chair
- pen or any solids around you

Questions

Use the questions below to investigate each solid.

Answer the following based on each solid

Stone

1. Does it feel hard or soft?
2. Does it make a sound when you knock on it?
3. Does it break easily?
4. Did you put your finger through it?
5. Was your hand dry or wet after handling the object?
6. Does it change its shape when you put it in something else?
7. How would you describe the shape? was it fixed? does it remain the same?

NOTE: The questions above may be used to investigate any other objects that are solids.

ACTIVITY

USE THE TABLE BELOW TO FILL IN SOME OF YOUR ANSWERS ABOUT EACH OF THE OBJECTS

There are some empty rows at the bottom for you to fill in any other solid objects that you might have investigated.

OBJECT	OBSERVATION
Stone	
Cloth	
Paper	
Wood	

QUESTIONS

1. Write down common properties for all the solids you investigated?
2. List some other solid objects in your classroom. Give at least 2 examples.

We have learned that a substance in a solid form will have a fixed shape and takes up a definite space. **Let's now look at liquids**

ACTIVITY: Exploring the properties of liquids

MATERIALS:

- water
- baby oil
- fruit juice
- methylated spirits
- 5 small pieces of cloth
- 5 containers for each of the 5 liquids
- 5 other clean and empty containers, such as a glass, cool drink bottle or tin
- 5 saucers

INSTRUCTIONS:

Study the different liquids on the table and answer the following questions.
Write your answers in the table that follows. **DO NOT TASTE THE LIQUID!**

Water

1. How does it smell?
2. Can you put your finger through it?
3. Can you soak the liquid up with a cloth?
4. Put a small amount of the liquid in the saucer and leave it for a while in a warm place.
Was it easy to pour the liquid from one container to another?
5. Can the liquid flow or spread out on a saucer?
6. How will you describe the shape of the liquid, is it fixed, does it take the shape of the container?
7. Did the amount of the different liquids remain the same after leaving them in a warm place?

Do the same with the other liquids at your disposal

WASH YOUR HANDS AFTER HANDLING THE LIQUID

OBSERVATION	ANSWER
What did your liquid smell like ?	
Did the shape of the liquid change when you poured it into another container?	
What do you think happened to the liquid when you left it in a warm place?	

QUESTIONS

1. Write down the safety rules for this investigation.
2. Write down the common properties for all the liquids investigated

After doing this activity where we investigated the properties of liquids, we can say that a liquid:

- can flow,
- has no fixed shape; and
- takes the shape of the container that it is in.

Now that we know the properties of solids and liquids, let us investigate gases.

Gas

When we breathe, we are taking in and giving out gases. But we cannot see the gas.

Most gases are invisible and there are few that are visible.

The gas used in a stove to cook food. The smoke coming out of the exhaust of a motor car has carbon monoxide gas. Oxygen gas for patients with breathing problems. The air you breathe in has oxygen gas. The air you breathe out has more carbon dioxide gas

ACTIVITY

Learning about gases from pictures

INSTRUCTIONS:

1. Study the pictures below. Each of the pictures shows a different property of a gas.
2. The properties are listed in the first column of the activity below. Decide which picture is showing the property and give it a tick. Make a tick under the right picture for each property.
3. The first one has been done for you.

Property illustrated			
A gas moves without something that pushes it - it diffuses through the air.		✓	
A gas has no definite shape and			

fills the container it is in.				
A gas can be pressed to fill a smaller space.				

In summary, gas

- takes up all the space available.
- has no definite shape.
- can flow.

Now that we have learnt about solids, liquids and gases, let's see if we can identify solids, liquids and gases

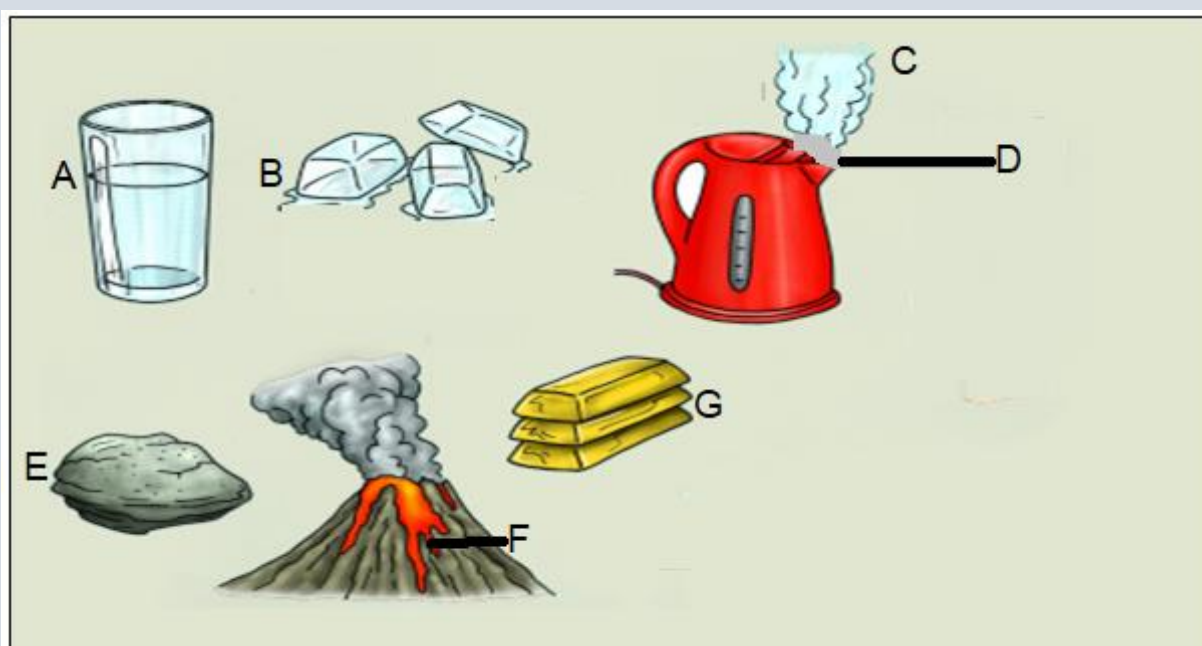
ACTIVITY

Identifying solids, liquids and gases

INSTRUCTIONS:

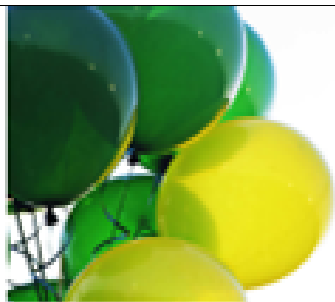
Look at the pictures of the different substances below and decide if they are solids, liquids or gases.

Use the table below place a tick in the right column.



SUBSTANCE	SOLID	LIQUID	GAS	
A. water in the glass				
B. Ice blocks				
C. Condensed water (cloud)				
D. Steam				
E. Rock				
F. Lava from volcano				
G. Gold bars				

Comparing solids, liquids and gases

SOLIDS	LIQUIDS	GASES
Have a definite shape	Have no definite shape	Have no definite shape
Takes up a definite space	Takes up a definite space	Takes up all the space available
Do not flow	Can flow	Can flow
		 <i>balloons filled with helium gas</i>

<https://cdn.shopify.com>

<https://www.bing.com>

1.2 CHANGE OF STATE

We learnt about the states of matter? These were solids liquids and gases. A substance can change from one state to another. For example, a solid can change into a liquid, water can be a liquid in your glass or turn into ice in the freezer. Ice is a solid. but what makes these substances change from one state to another?

Let's use water as an example.

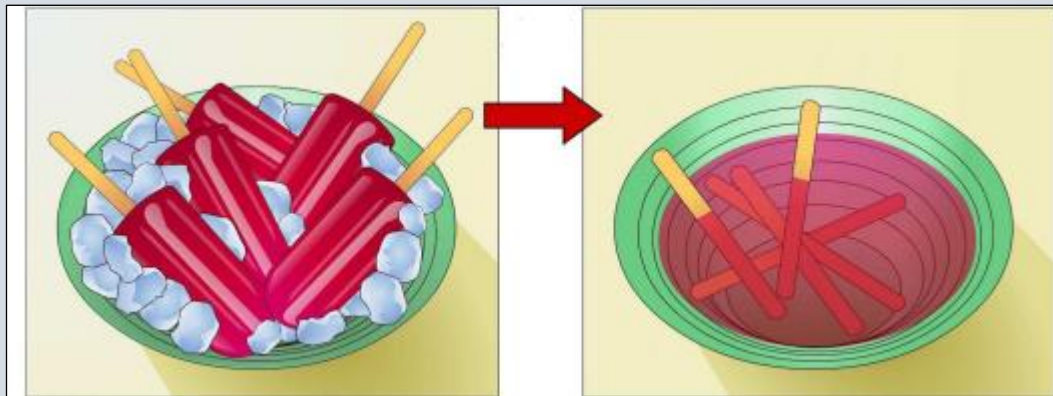
QUESTIONS

1. If you place tap water into an ice tray and put this in the freezer, what will happen to the water?
2. If you now take ice cubes and place them in the sun, what happens to the ice cubes?

If we place the water in a freezer, it freezes. If we place the ice cubes outside the freezer, they melt. This is because the state of matter can be changed from one to another by adding or removing heat.

ACTIVITY

Look at the picture of frozen ice lollies that were left in the sun and answer the questions



QUESTIONS

1. When the ice lollies are frozen, are they a solid, liquid or a gas?
2. What happened to the ice and the ice lollies while they were left in the sun?
3. What is the name given to this process?
4. Can the process of melting be reversed?

The ice lollies were frozen and cold. When they were placed in the sun, they started to warm up. This heat caused a state change to take place. The ice changed to a liquid. This is called **melting**.

These ice lollies were frozen, but they were made from a liquid. The liquid was poured into the shape of an ice lolly and then they were cooled as heat was removed and they froze. When a liquid changes to a solid, this is called **freezing/solidifying**.

Let's do some practical demonstrations in class to learn more.

ACTIVITY

Heating and cooling to cause a change of state

MATERIALS

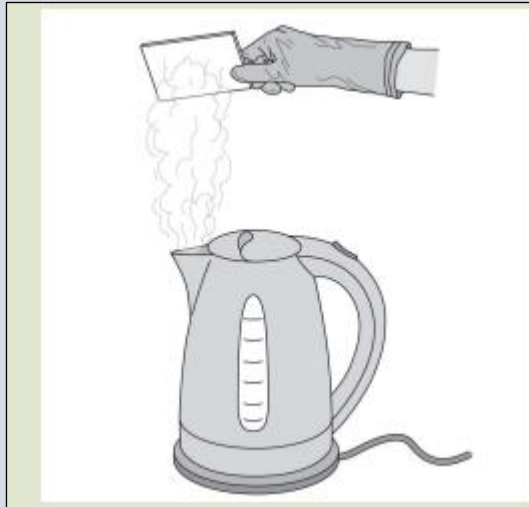
- kettle
- liquid water
- glass or mirror
- gloves or towel

INSTRUCTIONS:

Boil the water in the kettle.

Put a glass or mirror 30 cm above the boiling kettle (you need to wear gloves made of thick material or use a towel to avoid burning your skin)

This activity shows boiling and condensation, not necessarily evaporation.



1. What is the change of state when the water boiled and became steam?

The steam that comes out of the kettle is extremely hot and you cannot see it. The steam quickly cools and forms tiny droplets in the air. These tiny droplets are visible and form the "cloud" that you see. When these tiny droplets hit the mirror they cool more and form the bigger droplets which you see forming on the mirror. We say the steam condensed to form water. The change of state from the gas state to the liquid state is called **condensation**

When you leave a glass filled with cold water on the table, small droplets form on the outside. This is because there is water vapour in the air which cools down when it is near the cold glass. The water vapour in the air around the glass condenses as it changes from a gas to a liquid and forms the tiny droplets you can see.



Water droplets on the outside of a cold glass

Evaporation takes place when heat is added to the liquid. It means the water changes from the liquid to the gas state.



We hang wet clothes out to dry in the sun. They dry as the water evaporates.

ACTIVITY

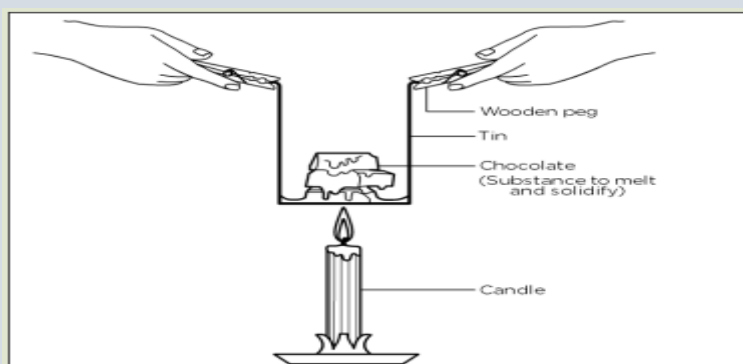
Melting and solidifying substances.

MATERIALS:

- butter, fat or margarine
- chocolate or wax
- ice blocks or ice cream
- 3 containers which will not melt (they can be empty tins)
- 6 wooden pegs
- a candle
- matches

INSTRUCTIONS:

1. Plan how you are going to melt and solidify the substances.
2. Look at the diagram below which shows how you can do this.
4. Test each different substance that you have by placing it in the tin and holding it over the candle.
5. Then remove the tin from the candle and leave it on the side to cool.
6. Observe what happens to each substance and write down your observations in the table below.



Setup for the investigation.

Substance	Observation before heating	What happened after heating	What happened after cooling
butter/margarine			
chocolate/wax			
Ice blocks/Ice cream			

QUESTIONS

1. What happened when the solids were heated by the candle?
2. What happened to the substances when they cooled down again?
3. What do we call the process when a liquid changes to a solid?

Here is a summary of the different state changes:

Change of state	Heating or cooling	We call the process
Solid to liquid	Heating	Melting
Liquid to gas	Heating	Evaporation
Gas to liquid	cooling	Condensation
Liquid to solid	cooling	Freezing and solidifying

REVISION

1. What are the three states of matter?
2. Describe what happens to solid ice when it is heated.
3. Below are the definitions of each of the three states of matter. They are placed in the wrong order. Match the right letter to the number and complete the last column.

Definition	State	Answer
3.1. Has a definite shape and takes up a definite space on the surface.	Liquid	
3.2. Has no definite shape and spreads in the space.	Solid	
3.3. Has no definite shape and takes the shape of a container.	Gas	

4. What will happen to the water in a saucer if we leave it in the sun for four hours on a very hot day?
5. Name 2 properties of solid
6. A block of ice, is left in the sun on a hot day. Discuss what changes you would observe in the object after three hours.
7. What is the reverse of freezing?

KEY CONCEPTS

- Matter is everything around us.
- Solids are matter that has a fixed shape.
- Liquids are matter that runs or flows, can be poured, takes the shape of the container.
- Gases are mostly invisible and takes the shape of the container and spreads out / flows in space.
- A change of state is brought about by heating and cooling matter.

TOPIC 2: METALS AND NON-METALS

KEY QUESTIONS

- How can we tell if something is made of a metal or a non-metal?
- How do we decide what material to use when we want to make or build something?

In this chapter we will learn about metals and non-metals. Metals and non-metals are two different classes of materials. Each class has its own unique properties.

DEFINITION OF CONCEPT

Metals

A shiny solid that can conduct heat or electricity and can be formed into sheets or wire
e.g. copper, iron, silver.

2.1 Properties of metals

- Metals are usually shiny.
- Most metals are hard and they feel heavy.
- Metals conduct electricity and heat well.
- Metals are malleable (they can be shaped into flat sheets) and
- They are ductile (this means they can be made into thin wire)

2.2. Properties of non-metals

Non-metals are materials that do not have the same properties as metals.

- Non-metals are not shiny, but tend to be dull.
- Many non-metals are not bendy (flexible) but brittle. This means that they will break when we try to bend them with enough force.
- Non-metals do not conduct electricity or heat well

ACTIVITY

What are the properties of metals and non-metals?

AIM: To describe the properties of metals and non-metals.

MATERIALS:

- coins
- metal spoon
- metal nail or screw
- steel wool
- cardboard
- cotton wool
- plastic spoon
- piece of chalk

METHOD:

1. Few metal objects have been placed in front of you. Notice all the different shapes. Write the name of each object in the table below.
2. Hold each object in your hand. Is it rough or smooth?
3. Look at each object carefully. Is it shiny or dull? Can you describe its colour?
4. Drop each object on the floor, or tap it. What sound does it make?
5. Dip the metal spoon and plastic spoon in a cup filled with hot water for 5 minutes. Feel the handle of the spoons. What do you feel? (Is it hot/cold)

6. OBSERVATIONS

Fill in the observations from your investigation of different metals and non-metals below.

Object	Shiny/dull	Malleable	Hardness	Heat conduction	Metal/non-metal

CONCLUSION

What have you learned from investigating the properties of metals and non-metals?

Distinguish between metals and non-metals

What have we learnt about the properties of metals and non-metals?

Now we are ready to compare the properties of metals and non-metals.

Metals	Non-metals
solid and strong.	Can be soft or flexible.
Malleable and ductile (this means they can be hammered or bent into different shapes).	Can be hard and brittle.
Shiny or silvery (lustrous).	Dull.
Cold to touch.	Usually feels neither cold nor hot.

ACTIVITY

Name 3 different materials in your immediate surroundings and describe their features

Material	Colour	Texture	Toughness

ACTIVITY

Research and describe how metals and non-metals are used to make products for example furniture made in the workshops at school.

2.3 USES OF METALS AND NON-METALS.

Metals and non-metals have thousands of uses. We use them every day, sometimes even without knowing!

Here are some of the uses of non-metals

- Oxygen is used for breathing
- Chlorine is used for purifying water
- Nitrogen is used in fertilizers
- Graphite is used as a good conductor of electricity

Here are some of the uses of metals

- Copper is used for making electrical wires, cooking utensils
- Iron or steel is used to construct the house and bridges
- Zinc is used to protect the iron from rusting
- Gold and silver are used for making jewellery

The pictures highlight some of the uses of metals



Aeroplanes are made from strong, durable sheets of metal.



A bridge made of metal.

KEY CONCEPTS

- Every type of matter has its own set of properties
- Metals and non-metals have different properties.
- Materials are useful because of their properties.
- Metals and non-metals have many uses.

REVISION

1. List 3 properties of metals.
2. Why do some pots and pans have handles made of plastic or wood?

TOPIC 3: PROCESSING MATERIALS

KEY QUESTIONS

- How can we make new materials?
- How does the amount of material we start with affect the amount of new material we can make?

This topic deals with the many different ways in which materials can be processed. The important messages are that new materials will have new properties, and that processing always has a purpose.

There are many ways to process existing materials into new materials. There are also many reasons why we would need to process materials into new materials. When we bake a cake, we are processing flour, eggs and other ingredients (that may not taste very nice on their own) into cake which tastes really good! We process materials to make them stronger, or more durable, or waterproof, or even just to make them look more beautiful or interesting. New materials that form after mixing different materials are sometimes called mixtures

3.1 DEFINITION OF CONCEPTS

- **Combining**
Mixing different materials to make new products
- **Mixing**
Combining of two or more materials to achieve a desired level of uniformity
- **Cooking**
Skill of preparing food by combining, mixing and heating ingredients
- **Heating**
Conversion of energy to heat
- **Cooling**
Removal of heat
- **Drying**
Removal of water by evaporation from a solid

Materials can be processed to make new materials or products.

When we combine materials, new materials are made. The properties of the new material are often different from the properties of the material we started with. There are many ways to process existing materials and many reasons why we would need to process materials into new materials. This might include materials that can mix and set, such as plaster and water, baking a cake or using jelly powder.

3.2. PROCESSED MATERIALS MADE BY PEOPLE IN AFRICA

Traditional processing.

- In the olden days only natural materials were available and people found many clever ways to make these materials more useful.
- They also had ways of reinforcing the mud used for making traditional huts.
- They also knew exactly which reeds would make the best mats to cover their walls, and how to build the best houses for their climate and lifestyle.
- Some of these traditional ways of processing materials are still used today.
- We will learn about a different kind of traditional home, which is still seen today.
- Some of the traditional homes in Africa are made of clay or mud.
- Mud (mixture of soil and water) . When it is reinforced it can make a strong and durable building material, which can be used to build a house. If it is well built, the house will last for many years.

ACTIVITY

Making a wall of a mud house

Study the photos. Then answer the questions.



A boy helping to build a mud wall.



The wall of a mud house with a stick frame.¹

1. What material is the house in the pictures made of?
2. Look at the close-up photo of a wall, to see how he strengthened the wall of his house. Name the method used to strengthen the walls.
3. List three reasons why we process materials
4. Below are two pictures. Describe the properties of the materials in both pictures and what processes took place to get from Picture 1 to Picture 2.

Picture 1



Picture 2



In Africa, many people make objects by using plant products, called plant fibres. The people weave and stitch the plant fibres together to make different objects, such as reed mats, baskets, or even thatch to make a roof for a house. This is also a type of traditional processing:



A woman weaving a grass basket.



A man weaving a reed mat.²

ACTIVITY

Identifying objects made from plant fibre

INSTRUCTIONS:

1. Each of the following photos shows an object made from plant fibre.
2. Identify what it is and how the people are using these objects.

OBJECT MADE FROM PLANT FIBRE	DESCRIPTION
	
	
	

3.2 METHODS OF PROCESSING MATERIALS

The main focus will be on **combining materials** and **how they can be processed to make new materials and products**. This might include materials that can mix and set, such as plaster and water. we will also look at baking a cake or using jelly powder in the classroom to demonstrate this process.

Combining materials

Process: To process material means to combine or mix it with other materials and/or to change it into something new by subjecting it to one or more processes (heating, cooling, firing, melting, etc.)

Let's look at more ways to combine and process materials.

ACTIVITY

Making glue (Mixing)

Often, when we mix materials together, the properties of the new material or product is different from the properties of the materials we started with.

Let's make some sticky glue to find out about this!

MATERIALS:

- flour
- water
- 2 bowls for the flour and water
- a bowl for mixing
- pieces of paper

INSTRUCTIONS

1. First you need to look at the individual properties of the flour and water before we mix them together. Describe the properties of the flour and water.
2. Now experiment with mixing different quantities of water and flour together to make a sticky paste.
3. See if you can stick pieces of paper together using the glue that you have made!
4. Describe the properties of the paste you have made.
5. Is the final product after processing a totally new product?

QUESTIONS

1. What did the flour feel like before it was mixed with the water?
2. How would you describe the properties of plain water?
3. After mixing the flour and water together, what are the properties of the paste that you end up with?
4. What state of matter is the flour before mixing?
5. What state of matter is the water before mixing?

Did you get any of the paste you made in this activity on your fingers? Perhaps it started to dry and become hard? Often when we combine materials together we have to let them set.

ACTIVITY

Making Jelly (Mixing, cooling and setting)

Have you ever tasted jelly? Jelly comes in many different colours and flavours. Which is your favourite? To make jelly, we must dissolve jelly powder in hot water. When the solution of jelly powder in water cools down, something very special happens: The solution sets, and turns into a delightfully wobbly, sweet treat! The jelly powder has been processed into something new! That is what the next activity is all about.

MATERIALS:

- a packet of jelly powder
- a bowl
- a measuring cup
- hot and cold water
- spoon for mixing

INSTRUCTIONS:

1. Read the instructions on the packet of jelly.
2. Pour the jelly powder onto the bowl.
3. Look carefully at the dry jelly powder. What does it look like?
4. Touch the jelly powder with your finger. What does it feel like?
5. Place a few grains of the jelly powder on your tongue. What does it taste like?
6. Write your findings in the table below.
7. Follow the instructions on the packet to make the jelly.
8. Cool the jelly until it sets.
9. Describe the properties of the prepared jelly in the same way that you did for the starting materials.
10. Is the final product after processing a totally new product?

Here are some words that you may find useful. You may also use your own words.

liquid, clear, powdery, sweet, sticky, transparent, wobbly, solid, gelatinous (jelly-like), slippery

TABLE OF OBSERVATION

Properties	Dry jelly Powder (before mixing)	Water (before mixing)	Prepared jelly (after it has set)
What does it look like?			
What does it feel like?			
What does it taste like?			

QUESTIONS

1. What materials did you start with?
2. What happened to the jelly powder when you mixed it with the water?
3. Why is the water a different colour?
4. How did the jelly mixture change when it cooled down?

Mixing and cooking

Cooking food is also a form of processing. Have you ever seen what a raw egg looks like? The same egg looks quite different when it is cooked. Notice how the egg white is transparent when it is raw, and white when it is cooked. When it is raw, the egg is runny, like liquid. When it is cooked, the egg is solid but soft like rubber or soft plastic.

Look at the pictures below:



source: <https://www.allrecipes.com>

Now that we know that materials can be processed, let us now explore the properties and uses of processed materials

3.3. PROCESSED MATERIALS

Special Properties of processed materials

Processed materials are strong, durable (long lasting), waterproof, different colors (interesting and attractive), fire and heat resistant and cost effective.

Plaster of Paris

- Made from gypsum
- White powder is mixed with water to form a paste that can be moulded into various shapes. When it dries it becomes very hard.
- It is water proof and fire resistant
- Used as plaster cast, dental moulds, art works, fill cracks

Concrete

- Made from cement, water, gravel and sand
- Is an important materials used in building many things: roads, pillars, pipes, bridges etc.
- It is strong and hard
- It is durable
- It is fire resistant
- It is water proof
- It can easily be moulded into any shape

Ceramics and glass

- Ceramics items include casserole dishes, plates and mugs etc.
- Glass include glass cups, jugs and windows etc.
- Very useful because they are water proof
- Can withstand high temperatures (heat resistant),
- We can use them for cooking.

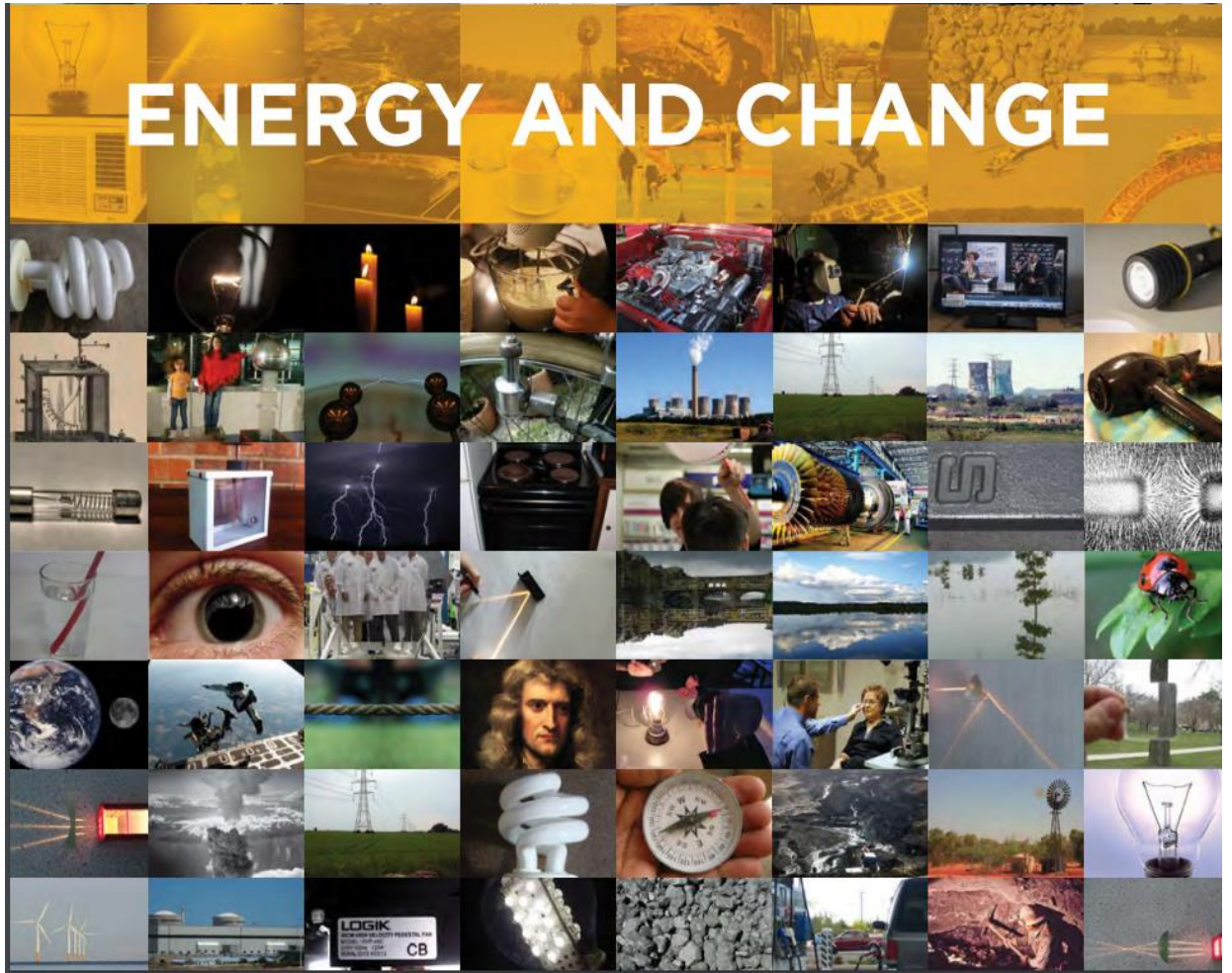
Fabric

- Made of fibres that are either natural or man made
- Natural fibres come from animals or plants e.g. Cotton from plant, wool from sheep, silk from silkworm
- Manmade fibres are made from other substances e.g. Nylon and polyester
- Wool jersey trap the warm air from our bodies which helps to keep us warm
- Cotton is cooler to wear.
- Towelling fabric is used to make towels and dish clothes.

KEY CONCEPTS

- Materials can be processed in many different ways to make new materials or products.
- When we process materials, the new materials may have different properties.
- The purpose of most processing methods is to make materials more useful.
- In Africa, people have processed materials for hundreds of years, for example to make clay pots and woven products.

TERM 3



CONTENT OVERVIEW TERM 3: ENERGY AND CHANGE

STRAND	YEAR 2	YEAR 3	YEAR 4
Energy and Change	<u>Energy</u> - Explain concepts with examples - Different forms/types of energy	<u>Energy and Temperature</u> - Explain the concepts with examples - Describe how to read and use a thermometer - Describe how heating and cooling can change materials	<u>Electric circuits</u> - Define concepts - Explain when a circuit is complete - Conductors and Insulators
	<u>Energy Sources</u> - Explain concepts with examples - Categorise sources of energy into Renewable and Non-Renewable sources - Name the different energy sources	<u>Stored energy in fuels</u> - Explain concepts with examples - Explain how energy is stored in food - Discuss the Renewable and Non-renewable sources	<u>Systems to solve problems</u> - Identify different circuits - Make simple electrical systems
	<u>Energy Transfer</u> - Explain where the energy from food comes from - Describe how energy is transferred in a food chain - Identify different types of energy transfer	<u>Burning Fuels</u> - Investigate, compare & describe how different fuels burn: Example candle <u>Safety with fire</u> - Identify sources - Identify safety rules when using fuels	<u>Mains electricity</u> - Describe concepts with examples - Compare the amount of energy released by different energy sources
	<u>Sound and Noise pollution</u> - Explain the concept sound with different examples - Explain sound as noise pollution and the impact on the environment	<u>Energy and Electricity</u> - Briefly define concepts - List the different types of batteries and its uses	<u>Saving Cost of Generating Electricity</u> - Explain concepts with examples - Describe energy sources and usage in South Africa - Compare different electrical appliances

ENERGY AND CHANGE CONTENT OUTLINE

WEEK	TOPIC	CONTENT
1-5	Energy and Energy Transfer	Energy from the Sun • Explain concepts with examples - Energy - Energy from the sun Energy around us Practical: Activity • Identify and demonstrate different types of energy transfer (examples) - electrical energy to: - heat energy - light energy - sound energy - movement energy - food energy to movement energy (Make use of a variety of examples) Doing this work practical is very effective.
6-7	Energy Sources	Energy Sources • Explain the following concepts with examples: - Energy sources (main source of energy- Sun, food, - Food as a source of energy - Food chain • Describe the organization of a simple food chain. - Energy from the Sun. - Producers (plants) - Users (Herbivore, Carnivores, Omnivores) - Decomposers - Renewable sources of energy e.g. Sun, wind and water. - Non-renewable sources of energy e.g. coal, crude oil, natural gas, fuels. • Discuss the classifications of both Renewable- and Non-Renewable sources of energy:

8	Sound and noise pollution	Noise pollution • Briefly explain the following concepts with examples: Sound; Noise pollution • Explain what noise pollution is and give practical examples in their environment. • Describe the danger of: - Loud sound - Sound to hearing
9-10	The weeks allocated for formal assessment are integrated across the term.	
Formal Assessment: • Practical: 50 % • Test (Pen and Paper): 50 %		

TOPIC 1: ENERGY

1. DEFINITION OF CONCEPTS

1.1. ENERGY

KEY QUESTIONS

- What is energy?
- What do we need energy for?
- Where does energy come from?

Energy

Energy is the ability to do work. Energy changes from one form to another



Energy from the Sun

We get almost all of our energy on Earth from the Sun. We call this energy **solar energy**. **Sol** means **Sun**.



Key facts

We need energy to carry out all our life processes, that we dealt with in term 1 Life and Living.

Without energy, there would be nothing:

No life, no movement, no light, no heat, and no people.

ACTIVITY: Energy is the ability to do work

Let us have some fun playing a game!



INSTRUCTIONS:

1. Divide into groups of six and go outside for 10 minutes to play "Hide and Seek".
2. These are the rules of the game:
 - a) One person needs to be the seeker.
 - b) The seeker needs to find a home - a tree or the wall will work well.
 - c) The seeker closes their eyes and counts to 20.
 - d) Everyone else must hide. They are the hiders.
 - e) When the seeker has counted to 20, (s)he must find the others and tag them by touching.
 - f) If the hiders can get to the seekers' home without him or her seeing them, they can block themselves and cannot be tagged by the seeker.
 - g) if the seeker finds a hider s/he runs to their home, touches it and says "1, 2, 3, Tag Emma!" (or whoever they saw).
 - h) If there is time left over, you can choose a new seeker and play again.
3. Return to class after 10 minutes and talk to in your group about how you feel.
4. Each group must choose a leader who will report back to the class.
5. Write down the words that are being used to describe how you are feeling after the game.
6. Write down some of the descriptive words below.

This game requires that you do a lot of running. You might even get tired from it. This means that energy is being used to do work.

ACTIVITY: Word search

INSTRUCTIONS:

1. Complete the word search by finding the words listed below.

Food, Energy, Work, Movement, Sun, Change, Light, Heat, Absorb, Reflect, Transfer, Chain

2. Circle them with a coloured pen or pencil.

R	S	U	N	Q	F	D	L	W	E	A	M
C	E	D	J	G	O	D	O	O	K	S	O
H	H	F	F	S	O	W	O	R	K	K	V
A	C	O	L	I	D	W	X	L	S	A	E
N	O	H	D	E	N	E	R	G	Y	S	M
G	F	G	A	I	C	W	J	U	S	N	E
E	V	V	X	I	B	T	Z	L	Q	K	N
H	E	X	Q	W	N	H	E	E	D	B	T
E	Q	V	C	T	R	A	N	S	F	E	R
A	E	Y	U	J	E	D	F	N	Y	Y	B
T	T	T	Z	U	T	K	L	I	G	H	T
K	E	P	A	B	S	O	R	B	H	U	M

1.2. Energy from the sun

Without the Sun, the Earth would be a cold place with no life. Energy from the Sun has many different **uses**.

- A. **Light and warmth:** We use the light from the Sun so that we can see during the day. We use the energy from the Sun to warm us.



People enjoying the Sun on the beach

- B. **Plants use light as energy from the Sun to grow.** Plants use light, water and air as requirements for growth. As the plant grows it stores some energy in its roots, leaves and fruit.



Plants use the energy from the Sun to make food, such as these mealie plants.

- C. **Animals eat plants to grow.** The energy stored in the plants is used by the animals for life processes.



These cows are eating grass to get their energy.

Plants get their energy from the Sun to grow. As it grows, the tree stores some of this energy in its parts. When we need heat and light at night and when it is cold, we burn the wood so that we can use the stored energy.

Long ago before dinosaurs lived on Earth, plants and animals also used the energy from the Sun to grow.



We burn wood to get warmth from the energy that is released.

Today some of these old dead plants and animals have turned into coal, oil and natural gas. Coal, oil and natural gas are called fossil fuels. We mine fossil fuels so that we can use the energy from the Sun that was stored millions of years ago.

ACTIVITY: Worksheet

Circle the picture that will need energy to work.



2.3. ENERGY AROUND US

- Energy is all around us.
- There are only two forms of energy namely: **Potential and Kinetic energy**

ACTIVITY: identifying energy around us

INSTRUCTIONS:

1. Look at the following picture.
2. Draw a circle around all the places where you think energy is being used.
3. If you have coloured pencils, you can colour the picture in.



QUESTION:

1. Write down five of the activities that you circled.

Light

Light comes from a light source. Anything that produces light is called a source of light.

- The Sun is a source of light.
- Stars are sources of light.
- A fire is a source of light.
- A candle is a source of light.
- An electric bulb is a source of light.

Heat

Do you enjoy standing outside on a warm summer day and feeling the warmth from the Sun on your skin? What about warming your hands on a frosty cold morning in front of a fire? You are feeling heat! We discussed that the Sun provides us with light, but it also provides us with heat.

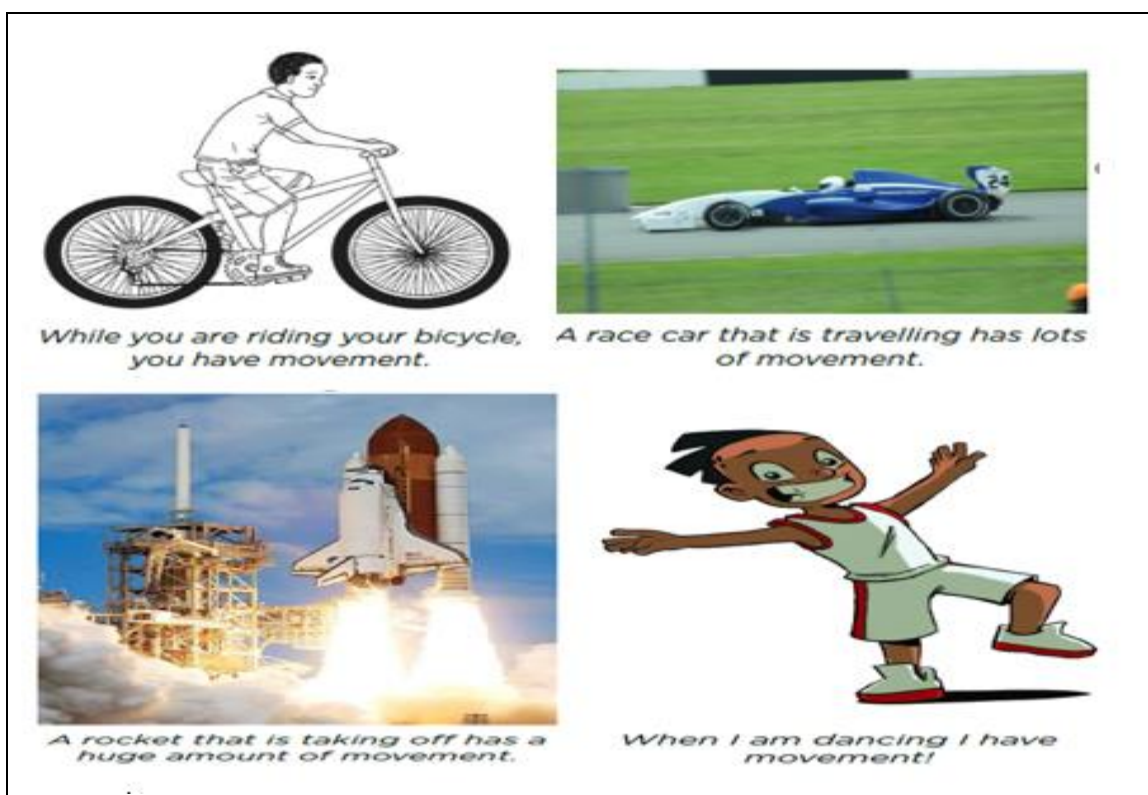
Look at these lions enjoying lying in the heat from the Sun!



Heat can be found in many different places. Anything that provides us with heat is a source of heat. Let's look more closely at different sources of heat energy.

Movement

When objects are moving, they have movement. The faster the object is moving; the more movement it has. Look at the examples of movement below.



Stored energy

Energy is also stored in some sources. Stored energy is the energy that is stored in our food, in petrol, in wood, oil and other chemicals. Batteries also contain chemicals which are used to store energy. This stored energy can be used for different purposes.

Energy can also be transferred (moved) from one part of a system to another part.

We use electricity every day in our modern lives. Electrical energy can be transferred from a source to an appliance or light bulb.

Look at this picture of the light bulb below. Electrical energy is transferred from a source to the light bulb to make it glow.



A light bulb gives off light energy and heat energy.²



In a torch, energy is transferred from the batteries to the bulb.

Have you ever used a torch before? How do you think the bulb lights up in the torch? The batteries are the source of energy in the torch. When the torch is turned on, the

energy is transferred from the batteries to the bulb to make it light up so you can see in the dark.

This brings us to the next section. We can think of the energy in the batteries of the torch as being the input energy and the light energy that is given off by the bulb as the output energy.

Sound

Did you know sound is also a type of energy?! Sound is everywhere.

ACTIVITY: Observing sound energy around us

INSTRUCTIONS:

1. Close your eyes and be very quiet for two minutes.
2. Be very still and listen to your surroundings.
3. Write down five different sounds that you heard.

Sound is a special type of movement. Sound travels as vibrations that we can hear and sometimes even feel. All sounds are caused by vibration of substances.

Sound can come from different sources. You can clap your hands, stamp your feet, talk, sing or play a musical instrument. Different vibrations will make different sounds.

ACTIVITY: Use a ruler to make sound

MATERIALS: (What you will need)

- a ruler
- the edge of a desk

INSTRUCTIONS (What to do):

1. Put the ruler on the table so that it sticks out over the edge.
2. Push the ruler down.
3. Pluck the ruler and hear the sound.



QUESTIONS:

1. You can see the vibrations of the ruler as it moves. Can you hear the vibrations?

ACTIVITY: investigating the input and output energy of appliances.

INSTRUCTIONS:

1. Look at each image and write down the type of output energy that it transfers to the environment.
2. Some of the appliances may transfer more than one type of energy to the surroundings.

Appliances	Energy transferred to surroundings
Stove 	
Kettle 	
Lamp 	
Vacuum cleaner 	
Electric fan 	
Drill 	

In summary, we can say that in appliances and machines, the input is normally electrical energy and the output depends on the appliance:

Input and output energy

Whenever anything happens, energy is transferred from one component into another. People, machines and appliances need an energy input to work. They also have an energy output that may be useful.

Let's look at some examples.

Example 1: A girl is running a race. In order for the girl to have energy, she needs energy from somewhere. Her input energy is the **chemical energy** from the food that she ate. By running the race, she is giving out energy in the form of **movement energy and heat**.



Example 2: A TV will only work if it is plugged in. It needs energy to work. While watching TV, **electrical energy** is the **input** and **light and sound** is the output.



Example 3: A torch will not work when you turn it on unless it has batteries. The input energy for the torch to work comes from the **chemical energy** in the batteries which is changed to **electrical energy**. The output energy from the torch is **light and heat energy**.



MACHINES AND APPLIANCES

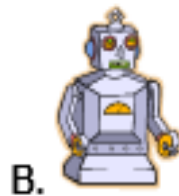
We use lots of appliances in our lives. These machines and appliances need input energy to make them work. This is usually electrical energy. The output energy (the work the appliance or machine does) is something that is useful to us.

ACTIVITY

Look at the picture below that describes the transfer of energy. Fill in the table with the correct letter representing the picture.



Light bulb



Robot



Iron



Radio

Energy can be transferred to _____	Item (picture)
1. Sound energy	
2. Heat energy	
3. Kinetic energy (movement)	
4. Light energy	

ACTIVITY

Indicate how stored energy is used in each of the pictures below:



Energy is stored in food, such as eggs, which we eat for breakfast.



Burning coal releases the stored energy as heat and light. ¹



Natural gas stores energy which is released as heat when it burns to cook our food.



When your parents or taxi driver turn on the ignition in their car, the energy stored in the petrol or diesel is released.

TOPIC 2: ENERGY SOURCES

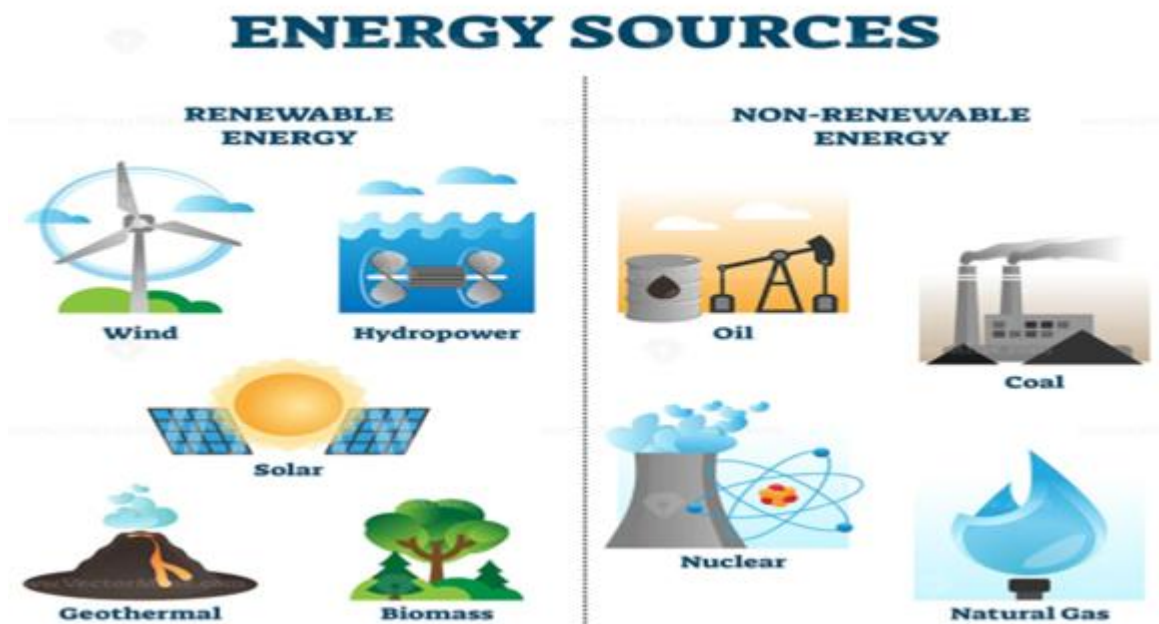
2.1. DEFINITION OF CONCEPTS

- 2.1.1. **Energy sources** have energy that is stored within them and can be used to make something happen, for example, energy stored in petrol can be used to make a car go.
- 2.1.2. **Renewable sources of energy** are ones which can be recycled or reused.
Examples are: Sun, Wind; Water,
- 2.1.3. **Non-renewable sources of energy** cannot be reused and so there is a limited amount available and when that runs out there will be none left.
Examples are: Coal, Nuclear energy, Natural gas, Petroleum,

2.2. RENEWABLE AND NON-RENEWABLE ENERGY SOURCES

We have two types of Energy resources, namely: **Renewable and Non-Renewable** Energy sources. **Renewable energy sources** referred to clean energy, comes from sources or processes that are constantly replenished. **Non – Renewable** are those once used, cannot be replaced. They will eventually run out.

Coal, oil and natural gas are formed below the surface of the earth. These fossil fuels are made from dead plants and animal remains and were deposited around the time when dinosaurs roamed the earth



ACTIVITY: Classification of energy sources

INSTRUCTIONS:

1. Study the following images which show different sources of energy.
2. Use the images to answer the questions that follow



Natural gas - gas burning on a stove top.



Oil - An oil rig sinks a drill into the ocean floor to reach the oil deposits



Sunlight - The Sun is a source of energy



Wood



Coal A coal mines



Hydropower - A large hydroelectric power station

Classify the sources of energy above into renewable and non-renewable sources of energy

RENEWABLE ENERGY SOURCES	NON-RENEWABLE ENERGY SOURCES

ACTIVITY 8:

1. Complete the following table

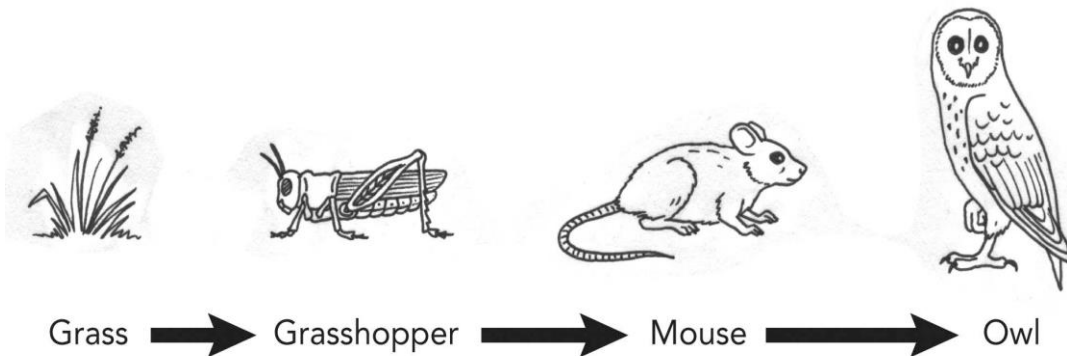
ENERGY SOURCE	WHERE IT IS FOUND	USES
1. Flowing Water	(A)	To generate electricity.
2.Wood	In trees / bush	(B)
3.(C)	In the sky	To provide energy.
4.Wind	(D)	To dry washing / to turn the wind
5. (E)	Underground / coal mines	To make fuel and fire.

2.4. ORGANIZATION OF THE SIMPLE FOOD CHAIN

The transfer of energy from the Sun to plants to people is called an **energy chain** or **food chain**. It is a chain because each organism forms a link in the chain as energy is passed along from one organism to the next.

The **arrows** show the direction of the **energy flow** from one thing to the next.

Look at the example of the food chain below.



Before reading or going through the next paragraph, or when you are explaining this concept to learners, first ask the question

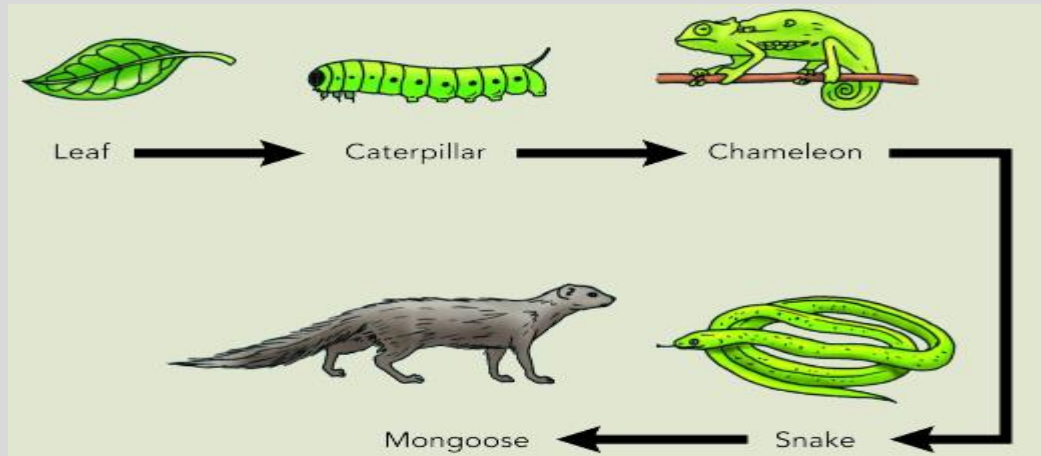
1. "What sort of energy does the Sun give off?"
2. Describe the transfer of energy from one organism to the next.

You can then go on to explain how the light energy given off by the Sun is used by the grass to make food and the energy is transferred from one organism to the next in the food chain. This ensures that you do not restrict the learners' view that the Sun gives off both light and heat energy.

ACTIVITY: Describing the transfer of energy from the sun in the food chain.

INSTRUCTIONS:

1. Look at the following food chain, describe the transfer of energy from one organism to the next.



1. In this food chain, what does the mongoose eat to get energy?

ACTIVITY:

Answer the following questions:

1. What does “energy is transferred means?”
2. For each of the following appliances, decide what is the input energy and identify the output energy.

Appliance	Input energy	Output energy
Radio		
hair dryer		
car moving		
riding a bicycle		
playing drums		
lights in your home		
plants growing		

REVISION:

1. in the table, fill in examples of different sources of energy

Light energy source	Heat energy source	Sound energy source	Stored energy source

2. Which sense organ do you use to detect sound?
3. A hearing-impaired person cannot hear music. How do you think someone who is hearing impaired could still dance to the music?
4. Which sense organ do you use to detect light?
5. What does "energy is transferred" mean?
6. List three things that contain stored energy that we can use.

TOPIC 3: SOUND AND NOISE POLLUTION

3.1. DEFINITION OF CONCEPTS

Sound

Sound is the type of energy made by movement.



Noise Pollution / Sound Pollution

Noise pollution is the disturbance of the environment with very loud uncoordinated sounds that are in most cases so irritating. **Noise Pollution** disrupt regular activities and reduce quality of life.



ACTIVITY:

PRACTICAL ACTIVITY: How do we make sound when we talk or sing?

INSTRUCTIONS:

1. Rest your hand gently on your neck and hum.
2. Make low humming sounds and higher humming sounds.
3. What do you hear?
4. What do you feel?

Examples of Noise /Sound Pollution

- Cars, buses, pedestrians, ambulances, and other street traffic noises
- Drilling or other heavy machinery in operation sounds like construction.
- Air traffic noise, such as planes taking off or landing, is always present at airports. Sounds from the workplace, which are frequently heard in open-plan offices.

ACTIVITY: Noise pollution



<https://www.bing.com/images>

1. List the examples of noise pollution in the picture.
2. Name any 4 noise pollution in your area or environment

DANGER OF LOUD NOISE OR SOUND POLLUTION

Problems related to noise include

stress related illnesses, high blood pressure, speech interference, hearing loss, sleep disruption, and lost productivity.

ACTIVITY: Making sound louder with a box

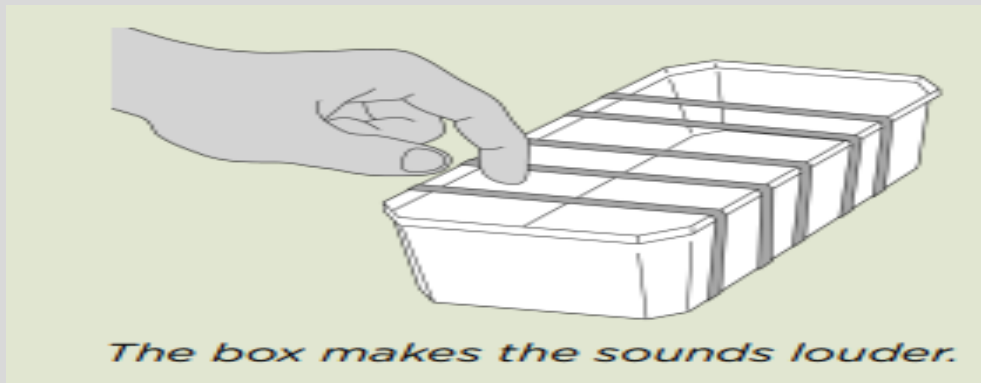
This activity can be combined with the elastic band activity earlier on.

MATERIALS:

- a cardboard shoe box or plastic box (a margarine tub or lunch box will work well)
- an elastic (rubber) band

INSTRUCTIONS:

1. Stretch the elastic between your hands and ask a friend to pluck it.
2. Now pull the elastic band over the box. The elastic must be tight. If it is not tight use a smaller elastic or a bigger box.
3. Pluck the elastic.
4. Feel the sides of the box as the elastic vibrates. Can you feel that the box is also vibrating? You may put a few rice grains inside to show the vibration of the box.



Let's summarise what we learnt from this activity:

- The box makes the sound seem louder.
- The box vibrates with the elastic.
- The vibrations of the box make sound inside the box.
- This makes the sound seem louder.
- Some musical instruments have a hollow shape.
- The sound is reflected inside the hollow.
- The walls of the hollow shape also vibrate as the sound
- echoes inside.
- This makes the sound seem louder.

ACTIVITY: Identifying noise pollution

QUESTIONS:

1. List five sounds that are very noisy, sounds that can hurt your ears, or just sounds that are difficult to live with.



2. Look at the picture of a learner in a very noisy environment. Identify all the sources of noise pollution.

TERM 4 CONTENT OVERVIEW

STRAND	YEAR 2	YEAR 3	YEAR 4
Planet Earth and Beyond	<u>Planet Earth</u> - Explain concepts with examples - Describe the features of the Earth - Describe the Earth as a planet in space	<u>Movements of the Earth and Planets</u> - Explain concepts with examples - Day and Night rotation - The Moon	Revision of term 1 to 3
	<u>The Sun</u> - Describe the unique features of the Sun - Describe and compare the size of the Sun to that of the Earth	<u>Systems for looking into Space</u> - Explain how people travel into space - Describe how telescopes are used to look into space - Describe the largest telescope build and used by South Africa	
	<u>The Moon</u> Describe the unique features of the moon		

CONTENT OUTLINE

WEEK	TOPIC	CONTENT
1-4	Planet Earth	Features of the Earth - Explain the following concepts with examples: Planet, Earth, , Sun, Star, Continent, Island, Habitats,, Moon - Describe the following features of the Earth: - Shape - Size - Movement - Ability to support life - Explain the difference between a continent and an island. - Identify the islands, oceans and continents on Earth – with their names. - Describe the different features of habitats on the Earth. Practical: Illustrate the Earth through a model or drawings - Describe the Earth as a planet in space. - Describe how the Earth, Moon and Sun moves in space. Discuss the different ways to observe space from the Earth
5-6	The Sun	- Describe and discuss the unique features of the Sun: - Size, Movement, Distance from the Earth, Temperature, Composition, How the Sun support life on Earth
7	Moon	Describe and discuss the unique features of the moon: Basic facts of the moon (Size, Distance from Earth etc.)
8 - 10	The weeks allocated for formal assessment are integrated across the term.	
Formal Assessment: - Practical: 40 % - Test (Pen and Paper): 60 %		

TERM 4: PLANET EARTH AND BEYOND

TOPIC 1: PLANET EARTH

KEY CONCEPTS

- A planet is a very big nearly round celestial (heavenly) body that goes around the Sun.
- The Earth is a planet. There are seven other planets moving around the Sun.
- The features of the Earth are the land with mountains and valleys, the water in dams, lakes, rivers and seas, and the air all around the Earth.
- Very big areas of land are called continents, and areas of land with water all around them are called islands.
- Living things grow on the land and in the water. The Earth has many habitats for many different living things.
- If we go up high above the Earth, everything looks different to the way it does when we are on the ground.

Key Questions

- What is a planet?
- What is planet Earth?
- What is the difference between a continent and an island?
- Are an ocean and a sea the same thing?

1.1. The Earth as a planet

A planet is a very big nearly round celestial (heavenly) body that goes around the Sun. Sun is a star that provides heat and light energy on Earth. Planets do not make their own light; they reflect light from the Sun. The Earth is one of the seven planets that are moving around the Sun. The other planets are very far from the Earth but stars are much further away.



Planet Earth as viewed from space

1.2. Features of the Earth's surface

Earth is the place where all people live. Earth is spherical (round) like a soccer ball and is made of rocks and water. The ground under your feet is part of the Earth. We live on the surface of the Earth. The Earth is covered with a layer of air called atmosphere.

Miners can dig deep tunnels under the surface of the Earth. The features of the Earth are the land, rocks and soil which makes up the mountains, hills and valleys, the water in dams, lakes, rivers and seas. Plants grow in soil. Look at the following pictures showing the different features of the Earth's surface.



Orange river



There are lots of rocks in this landscape



*The rocky sea shore on the Garden Route
showing steep cliffs, rocks and the ocean*



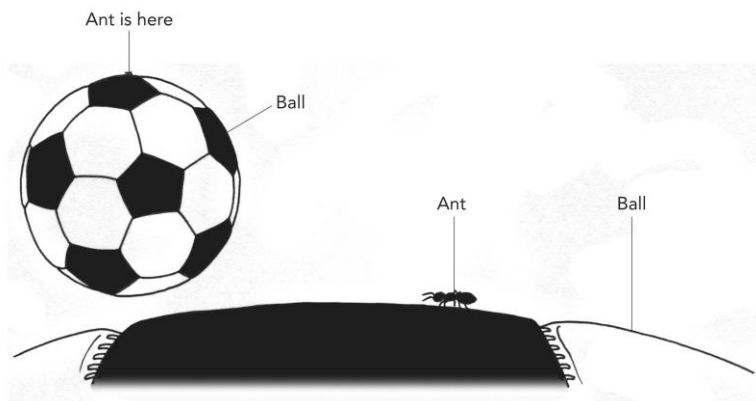
A deep valley between the mountains

1.3. The shape of the Earth

The Earth is shaped like a soccer ball and something that is shaped like a soccer ball is called a sphere. We see the shape of Earth when we go very high and get far away from it, for example an astronaut in a spaceship can see the shape of Earth when looking out the window of the spaceship and back at Earth.

If we look out of the classroom window, the Earth looks flat, not ball-shaped. Many people long ago believed that the Earth is flat, because it looks flat. It is so big that we can't see that it is curved.

Pretend you are an ant on a soccer ball. You are so small that the ball looks flat. You can't see the other side of the ball, and you can't see that it is a ball. Look at the picture of the ant on the soccer ball. All the ant can see is a flat surface. It does not even know he is on a round ball because the ball is so much bigger than it is.



The ant on the ball sees a flat surface as it is so small compared to the ball.

This is the same as us on Earth. We are so small compared to the Earth that when we are standing on the surface, the Earth looks flat to us. We cannot see that the Earth is actually round unless we look at pictures of the Earth taken from space.

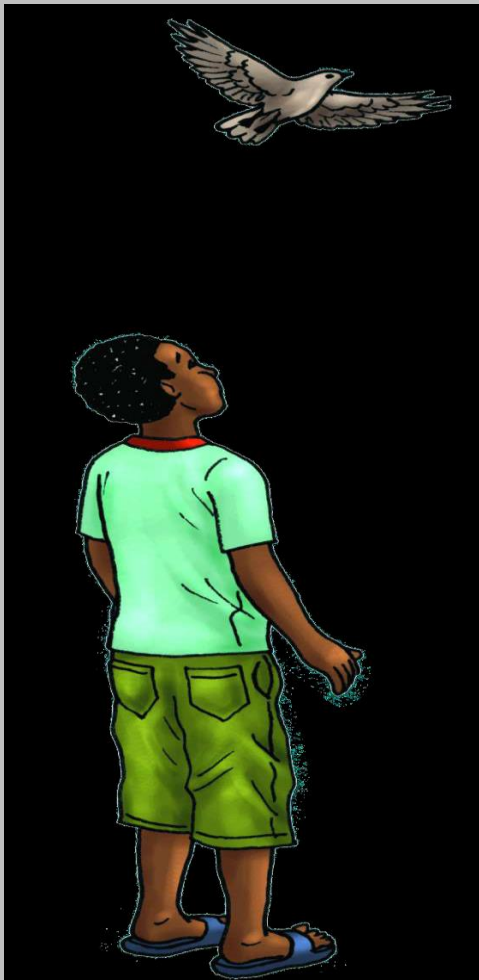
ACTIVITY: What do things look like from above?

MATERIALS:

- pencil
- coloured pencils
- rubber

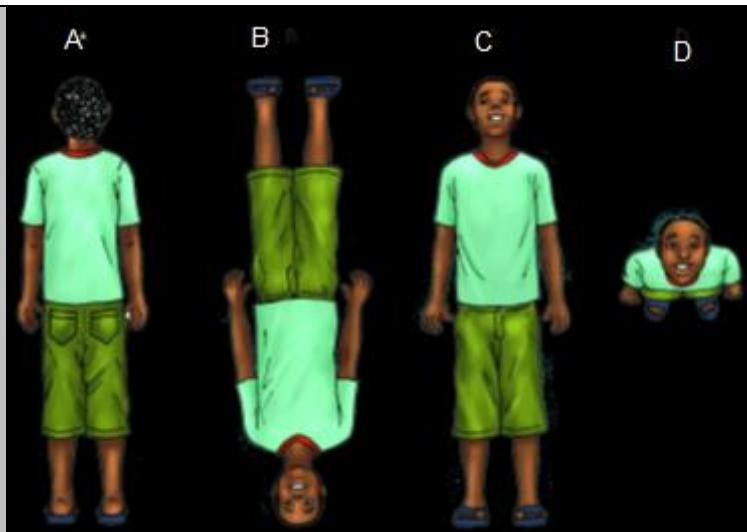
INSTRUCTIONS:

1. Let's look at what a bird sees when he flies over a boy. Can you see the bird flying over the boy in the picture below?
2. Answer the questions that follow

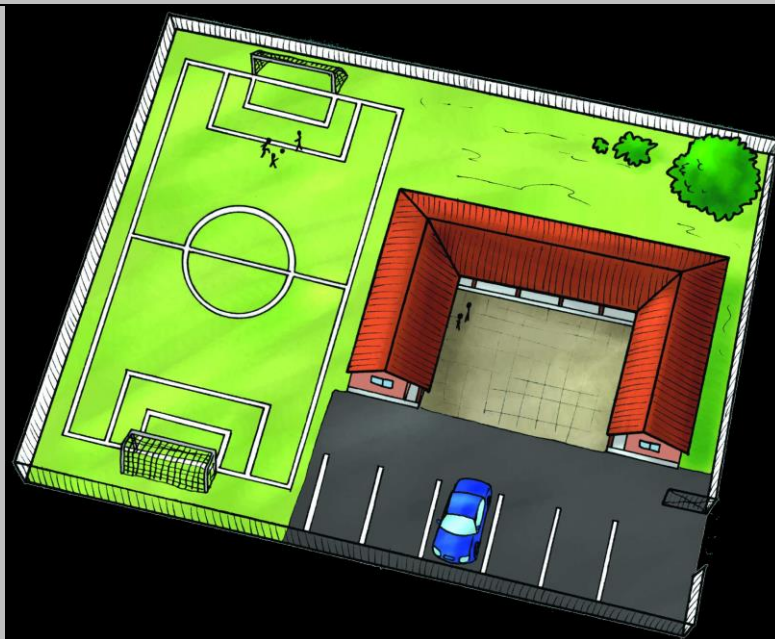


QUESTIONS:

1. When the bird looks down, what does it see? Choose the right picture from (a), (b), (c) or (d) by drawing a circle around it.



2. Imagine you are a fly on the ceiling in the classroom. You look down and see the classroom. Now draw the classroom as the fly sees it using the space below. Draw the chalkboard, the cupboard, the door and the teacher's table. You don't have to draw the people. You can use some colour if you want to.
3. The next picture shows you a school as it looks to a bird flying over.



The school yard as seen by a bird

4. Labels have been given below for some objects in the picture. Draw the letters A, B, C and D onto the picture to label the correct objects.
- A is the gate
 - B is the roof of the school
 - C is the soccer field
 - D is the tree

-
- A top-down view of a town map. The map features a green soccer field with a central circle and two goal areas. To the right of the field is a red building with a grid-patterned interior. Below the red building is a parking lot with a blue car. A road runs horizontally across the middle of the map, with a yellow truck and a green car on it. To the left of the road is a blue building. The map is bordered by a grey road on the right and a grey road on the bottom.

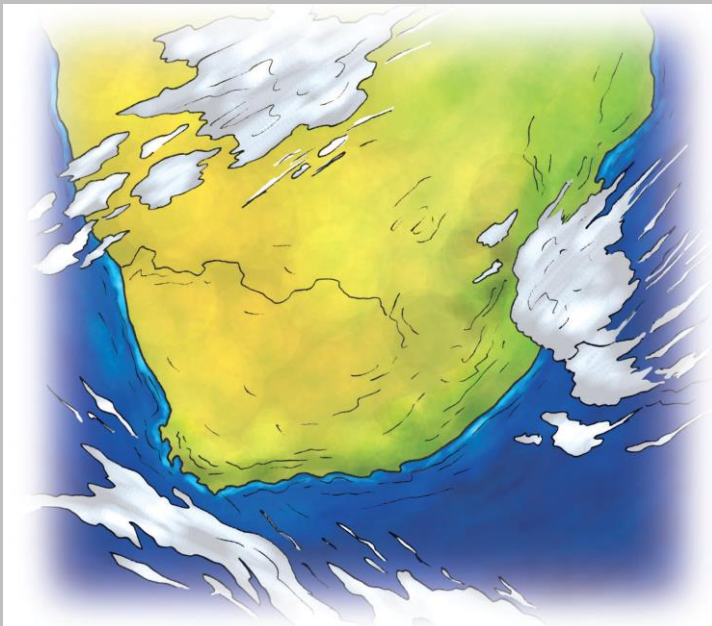
The school yard as seen by people in a low flying aeroplane

- ## YEAR 2 LEARNES' GUIDE



The town as seen by people in an aeroplane flying high overhead

What we have seen in this activity is that as you go higher and higher up, the objects appear smaller and smaller. So, when photographs are taken from an aeroplane or helicopter, we get a better idea of the features of the Earth's surface and we can see more. If we go even higher up and up and up into space, then we can't go in an aeroplane anymore. Then astronauts fly up in a spacecraft. In the picture below you see what South Africa looks like to people in a spacecraft high above South Africa.



This is what people in a spacecraft see when looking at South Africa from space.

1.4. The size and movement of the Earth

KEY CONCEPTS

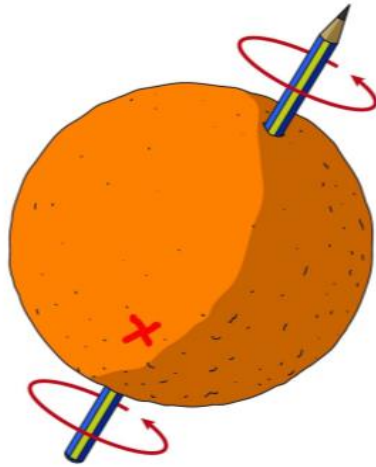
- The Earth spins on its axis. This is the reason we have day and night.
- The Earth also moves through Space, around the Sun.
- The Earth's path through Space around the Sun is called its orbit.

Earth moves in two different ways. It moves to make a day and it also moves to make a year. The Earth spins on its own axis. A complete spin on the Earth's axis makes a day. It takes the Earth 24 hours to complete a spin on its axis. A day has **daytime** (when there is sunlight) and **night-time** (when there is no visible sunlight). The size of the Earth is determined by its diameter which is 12 756 kilometres.



The Earth has an axis from the North to the South pole

Imagine an orange with a pencil stuck through it as in the picture below. The pencil is the axis of the orange. The Earth does not really have a pencil through it, but it does spin around like this orange on the pencil.



*The Earth is like the orange and the pencil is like the axis.
The curved arrows show which way the Earth spins.*

We are on the Earth. Let us imagine we are at the point where you see the red "X" on the orange:

- The Sun shines on the Earth and so we, at **X**, see the Sun. We call that daytime.
- But the Earth never stops spinning. So we, at **X**, move around with the Earth into the shaded part of the Earth. Then we cannot see the Sun any longer then it is night-time for us at **X**.

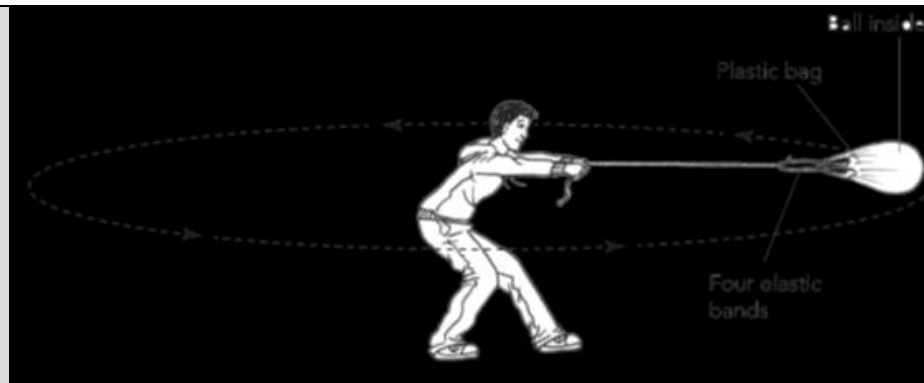
ACTIVITY: Make a model of the Earth moving around the Sun

MATERIALS:

- strong string about 5 metres long
- a ball in a plastic bag
- four thick rubber bands

INSTRUCTIONS:

1. Join the bag to the string, with four rubber bands.
2. Someone must run in a circle with the ball in the bag to help you get it going.
3. Then swing the ball around as fast as you can, on the end of the string. The rest of the class must watch the plastic bag carefully to see whether the rubber bands stretch.
4. You see the learner swinging the ball around himself. The learner represents the Sun and the ball represents the Earth. If you look carefully at the rubber bands, you can see that the ball is pulling on the learner, and the learner is pulling on the ball.



The ball orbits in a circle path around the learner

5. Take turns to swing the ball; feel how hard you need to pull on the ball to keep it going around.

QUESTIONS:

1. What do you feel as you swing the ball?
2. If the ball in its bag could feel, what would it feel?
3. If the string breaks, in what direction will the ball carry on travelling? Point with your hand to show your answer.
4. Why can you not see a circle in the picture above? Does the ball really move in a circle when you swing it?
5. The ball represents the Earth. You swing it quite fast, but how long does the Earth really need to go once around the Sun?

Now in Space, the Earth keeps on going around the Sun but there is no string pulling on the Earth. So, what pulls on the Earth? The force gravity of the Sun keeps the Earth moving in its orbit around the Sun.

1.5. Rotation and Revolution

The Sun seems to move from the East to the West. The Sun seems to come up (rise) in the East, move across the sky during the day and go down (set) in the West. The Sun does not actually move, it is the Earth that is spinning and that movement is called **rotation**.

The planet Earth orbits around the Sun in $365\frac{1}{4}$ days and we call that one year. As Earth moves to new positions around the Sun, the movement is called **revolution**.

ACTIVITY: Who is having daytime?

What you need:

- A globe of the Earth
- The photographs of the Earth labelled Picture A and Picture B.



A globe is a model of the Earth

INSTRUCTIONS:

1. There are two images of the Earth.
2. Look carefully at these pictures and use them and the globe to answer the questions.



Picture A



Picture B

QUESTIONS

Picture A

1. You are in South Africa. Find South Africa on the globe.
2. Find South Africa in **Picture A**.
3. Was it daytime in South Africa, when the spacecraft took the photo?
4. Was it daytime in Saudi Arabia? Hint: Use your globe to find Saudi Arabia and then find it in **Picture A** to see if it is day or night.
5. Was it day time in Argentina when this photograph was taken? Use the globe to help you locate Argentina. Explain your answer.
6. Now use the globe to locate Brazil. In Picture A, the tip of Brazil is in the sunlight. Is it morning or afternoon in Brazil? Why?
- 7.

Picture B

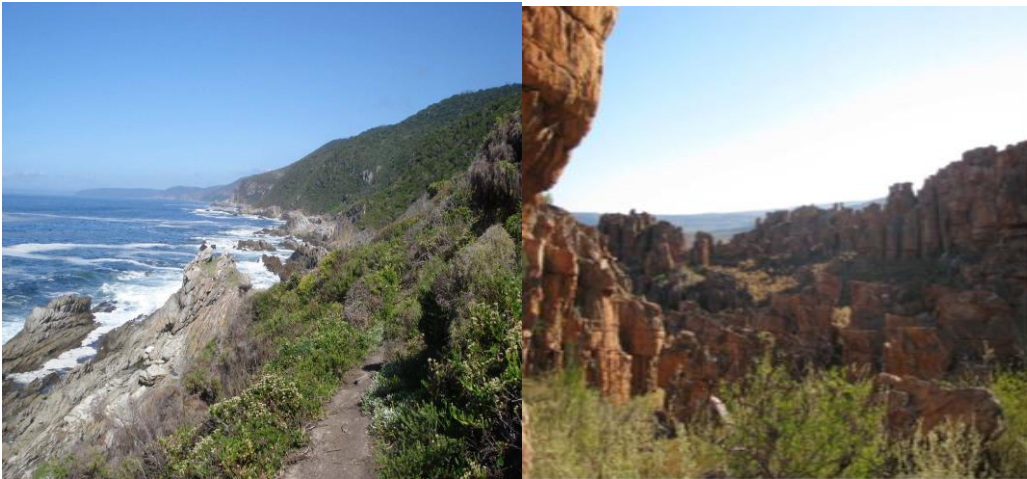
1. Look at Picture B. What part of the Earth is this picture showing?
2. Can you see the light on in Italy? When it is dark in Italy, is it still light in Spain?
3. Is it late afternoon or morning in Spain in Picture B?

1.6. Soil, air, water and sunlight support life on Earth

Life on Earth exists on the very thin layer around the planet - the crust. The soil is a thin layer that forms the top part of the crust. Plants need soil to grow in. The plants also use energy from the Sun to grow, and they make the oxygen. we and all the animals need to breathe.

1.7. Habitats

A habitat is a place where living organisms (plants and animals) live so they can find food, water, shelter and have their babies. The Earth has many habitats for many different living things. The different features of the Earth above are a few examples that provides habitats for various plants and animals.



The rocky sea shore on the Garden Route There are lots of rocks in this landscape showing steep cliffs, rocks and the ocean

ACTIVITY: Habitats on Earth**INSTRUCTIONS:**

1. Look at the pictures again on the pages before this activity showing different features of the Earth's surface.
2. Answer the questions below.

QUESTIONS:

1. Name some of the living things you can find on the Earth.
2. In the pictures, where do the birds live?.

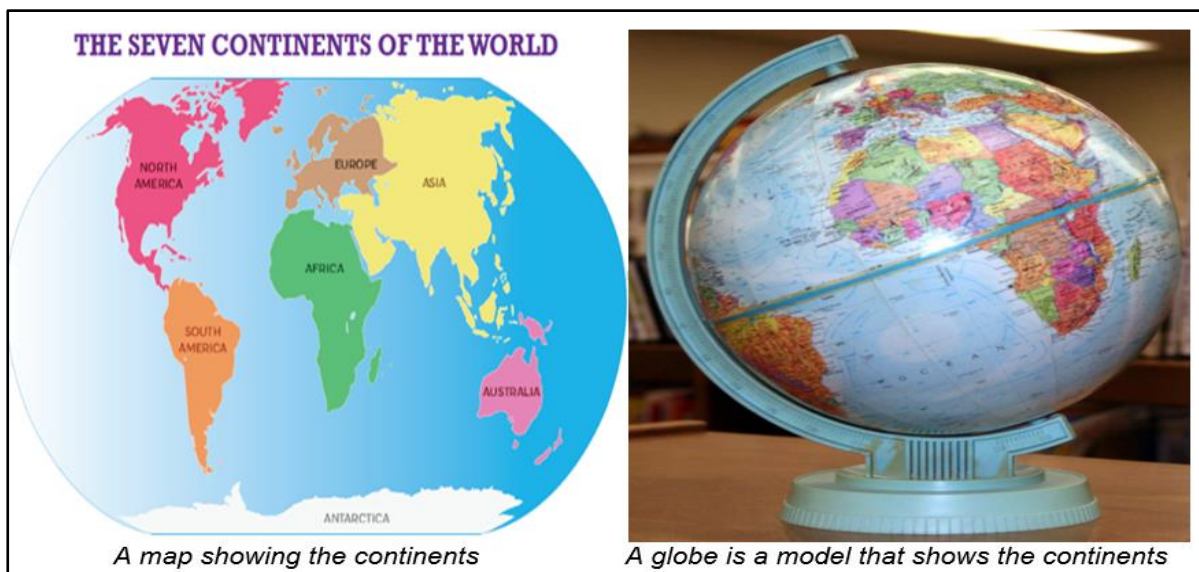
3. In the pictures, where do fish live?
4. Where can cattle live?
5. What kind of animals can live in deserts? A desert is a place with almost no water.
6. What kinds of animals live in forests?
7. Habitats have unique features, such as the rocky shore habitat which has crashing waves and big rocks. Name four habitats that you have seen in the pictures.
8. Name four non-living things you can see in the picture.

1.8. Continents

A continent is one big piece of land on the Earth. A continent has many countries. Africa is a continent with more than 50 countries.

The Earth has seven continents; they are:

1. Africa
2. North America
3. South America
4. Asia
5. Europe
6. Australia
7. Antarctica



1.9. Island

An island is a piece of land that is completely surrounded by water. Robben Island is a small island in South Africa. Madagascar, Zanzibar, Cape Verde and Mauritius are some of the African countries that are islands.

1.9.1. Difference between the Continent and Island

A continent is a big piece of land made up of many countries. An island is a small piece of land that is surrounded by water. It is either part of a country or only one country.

1.10. Oceans and Seas

Most of the surface of the Earth is covered with water (oceans and seas). This is why it is called the Blue Planet. Many people think that an 'ocean' and 'sea' is the same thing.

An ocean is a very large mass of water which covers a huge part of the Earth's surface. A sea is much smaller than the ocean and a sea is normally surrounded by land on some sides.



Oceans and seas of the world

TOPIC 2: THE SUN

KEY QUESTIONS

What is the Sun made of?

How far away is the Sun from the Earth?

The Sun looks so small in the sky, so how can it be bigger than the Earth?

Why is the Sun so important to us and all living things on Earth?

2.1. Stars

We see many stars at night. Many of those stars are very much bigger than the Sun. They look small because they are so very far away. Stars are balls of very hot gas and they make their own light. We can see many more stars through a telescope



Hundreds of thousands of stars seen through the Hubble telescope

Stars that look orange-reddish are not as hot as the Sun, and stars that look blue-white are much hotter than the Sun.

2.2. The Sun as a star

The Sun is the only star at the centre of our Solar System. Solar means Sun, system is therefore, Solar System means system of the Sun. The Solar System refers to the Sun and eight planets that move around it.

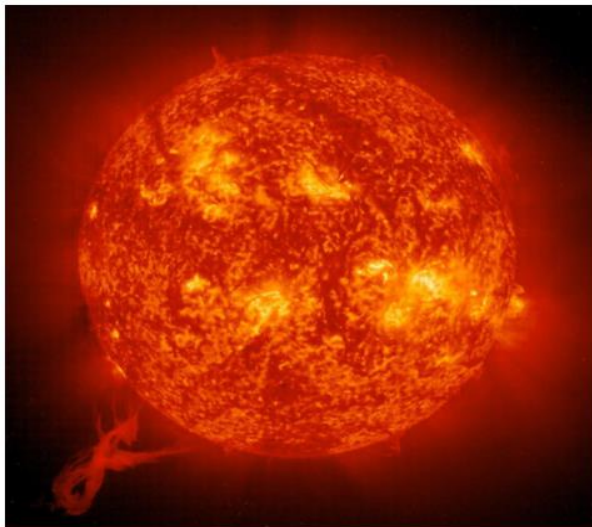
The Sun is a brightest star that is not moving while the planets move around the Sun in their orbits, as shown in the diagram below. The Sun and all the planets pull on each other as the planets move around it.

2.2.1. The Sun is the closest star

Why does the Sun look so much bigger than the other twinkling stars?

This is because the Sun is the closest star to us on Earth. The other stars in the sky are much, much further away.

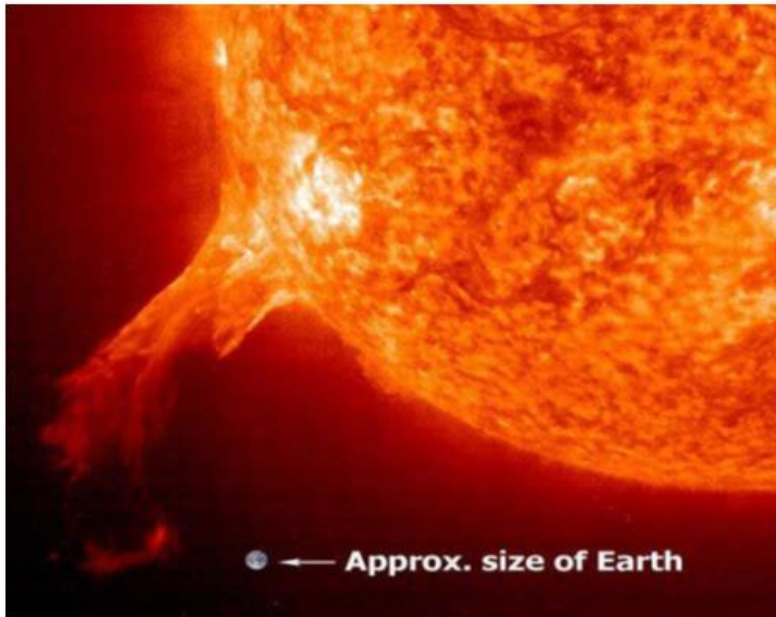
The Sun is a very big ball of hydrogen gas that is changing into helium gas, all the time, and this change gives off energy which makes the Sun very hot. It is nearly 150 million kilometres away from the Earth. The Sun is extremely hot - nearly 15 million degrees Celsius at the centre and 5 500 degrees Celsius on the surface.



*The Sun looks like this through a special camera.
Never look at the Sun with just your eyes.*

2.2.2. The Sun is very much bigger than the Earth

The Sun may look smaller than the Earth up in the sky. But this is actually because it is very, very far away. The Sun is much bigger than the Earth, it is so big that 1 300 000 Earths can fit inside the Sun. The Sun is way bigger than the Earth. The Sun's diameter is 1 392 000 kilometres and the Earth's diameter is 12 756 kilometres.



*This is how the size of the Earth compares to the size of the Sun.
The Earth is not really this close to the Sun.*

ACTIVITY: How big is the Sun, compared to the Earth?

MATERIALS:

- a piece of pavement
- a ruler or tape measure
- a piece of string longer than 60 cm
- a piece of chalk

INSTRUCTIONS:

1. Draw a circle 1 cm in diameter on the pavement with the piece of chalk. This represents the Earth.
2. Now, move about 1,5 to 2 metres away from the little Earth and give yourself enough space to draw the Sun.
3. Draw a circle 108 cm in diameter – this represents the Sun.
4. **To draw a circle with a diameter of 108 cm**, tie a string around the chalk. Measure 54 cm of string from the chalk and tie a knot there. Hold the knot on the board and move the chalk around the knot to draw a circle.

QUESTIONS:

1. If the Sun is so much bigger than the Earth, why does it look so small to us?

The pavement drawing shows the Earth quite close to the Sun. The Earth is actually very far from the Sun. It is 150 million kilometres from the Earth. That is 150 000 000 kilometres.

That is a really long way from the Earth to the Sun. If you went in a car at freeway speed of 120 km/h, you would have to travel for 146 years to reach the Sun. So, the Sun is far away and it is very big and very hot.

If the Sun is so far away, why does it look as big as it does? The Sun is so big that it is difficult for us to understand how big it really is. A model can help us to understand.

ACTIVITY: Use a model to show how far the Earth is from the Sun.

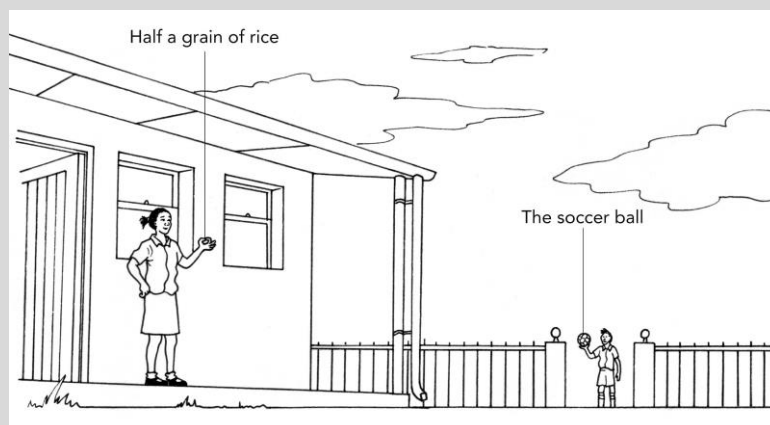
MATERIALS:

- a grain of rice broken in half
- a soccer ball
- space to move, such as the playground or soccer field

INSTRUCTIONS:

1. Look at the picture below.
2. One learner stands in a place where there is a lot of room all around them. This learner holds the soccer ball; this ball represents the Sun.
3. Another learner stands next to her and holds the half-grain of rice. It represents the Earth.
4. The learner holding the rice walks away from the learner holding the soccer ball with 24 of the biggest steps he can make. That distance is about 24 metres. The 24 metres represents the distance from the Sun to the Earth.
5. Now the learner carries the half-grain of rice, walking to walk to his right side. He must always stay 24 metres from the soccer ball. If he does this, he will walk in a circle around the soccer ball.

This model shows us that the small Earth moves in a circle around the big Sun.



The soccer ball represents the Sun, and the half-grain of rice represents the Earth.

QUESTIONS:

1. Stand 24 metres away from the soccer ball. Hold up one finger in front of you, and cover the soccer ball with your nail. Is the soccer ball really as big as your fingernail?
2. Why does the ball look as big as your nail?

2.2.3. The Sun is important to life on Earth

Without the Sun, life on Earth would not be possible. It would be completely dark and freezing cold. The Sun provides us with light and heat for the life to exist on Earth. Because of this light and heat, many other things become possible.

The Earth is the only planet in our Solar System with the right temperature for us to live; it is not too hot and not too cold. It is at exactly the right distance from the Sun, and has the suitable atmosphere for the perfect temperature to support life.

TOPIC 3: THE MOON

ACTIVITY: Investigate the Moon

INSTRUCTIONS:

You must find information out about the Moon.
Answer the questions below.

QUESTIONS:

1. How far is the Moon from the Earth?
2. Does the Moon have air for you to breathe?
3. Is there air between the Earth and the Moon?
4. Could a bird fly from the Earth to the Moon? Give a reason for your answer.
5. Can a big aeroplane fly to the Moon? Give a reason for your answer.

KEY QUESTIONS

- Why is the Moon bright on some nights?
- Why does the Moon seem as if it is changing its shape?
- How can we find out what the surface of the Moon is like?

3.1. Features of the Moon

The Moon is a rocky body orbiting planet Earth. It is smaller than the Earth, the Moon's surface is made of rock and grey sand. It is not a planet but a natural satellite of the Earth. A satellite is a body that orbits around another body in Space. The average distance between the Earth and the

Moon is 384 400km. Like the planets, it produces no light of its own, but we can see it because it reflects the light from the Sun.

The Moon orbits around the Earth, it appears as though the Moon is changing its shape in the sky. The illuminated shape that we see depends on where the moon is in its orbit around the Earth. Illuminated objects are objects that are able to reflect light in our eyes for example our Sun is an example of a luminous object, while the moon is an illuminated object.

The Moon rotates on its own axis, the same as its orbital period around the Earth and that is the reason we always see the same side of the Moon. The surface of the moon is covered with holes called craters.

These craters are made by space rocks that hit the Moon. They travel so fast that they make a round hole. The moon has no atmosphere, so there is no air to breathe, no wind to blow the dust, no weather and no water, so there is no rain to wash away sand. That is the reason why we see the craters on the Moon so clearly.

The Moon is the only place in our Solar System, other than the Earth, where humans have been. Moon can be viewed through binoculars. We see the moon both during the day and at night.



The Moon

This is the Earth seen from the surface of the Moon.

QUESTIONS

1. What made the craters on the surface of the Moon?
2. The surface of the Moon has many craters but the surface of the Earth has very few craters. Explain why that is so.
3. Neil Armstrong was the first man to put his foot on the Moon. His footprint is still there after 40 years. On Earth, a footprint does not last so long. Explain why it lasts so long on the Moon.

REVISION: PLANET EARTH

1. A person's nose and eyes and mouth are features of his or her face. Name four features of the Earth.
2. The earth is shaped like a ball. People do not fall down off the ball. What is the reason for that?
3. List the seven continents of the Earth.
4. Name an island that is part of South Africa.
5. Complete these sentences. Write out the whole sentence on the open lines, and use some of the words in the word-box to complete your sentence.

Word box: • water • continent • Blue Planet • one part • clouds

- a. The planet Earth looks blue and white from space. It is called the because _____ it is covered with _____ and _____
 - b. A continent is a large piece of land. For example, Africa is a _____ and South Africa is just _____ of Africa.
6. How many hours are there in a day?
 7. How many hours pass from sunrise until the next sunrise?
 8. How many days pass between your 10th birthday and your 11th birthday?
 9. How many times must the Earth spin around between your birthdays?
 10. Write out the whole paragraph and complete it using some of the words/phrases in the word box. You do not need all the words/phrases.

REVISION: THE SUN

1. Write out these sentences and complete them. Choose some of the words from the word box to complete the sentences. Write out the whole sentence.

Word box: (You do not need to use all the words)

- hydrogen gas
- helium gas
- plants
- light
- heat
- half a grain of rice
- Sun
- a soccer ball
- the Moon

When we compare the size of the Earth to the size of the _____; the Earth is the size of _____ compared to the size of a soccer ball.

The Sun gives _____ and _____ to the Earth.

All _____ need light and heat.

The Sun is not burning like a wood fire. The Sun is hot because _____ is changing into; _____.

REVISION: The Moon, Earth and the Sun

1. Complete the sentences using all the words from the word box. Write the sentences out in full on the lines below and number each sentence.
 - a) The Sun stays in _____.
 - b) The Earth rotates on its own _____.
 - c) The Moon revolves _____.
 - d) The _____ revolve together in a big circle around the Sun.
 - e) We only ever see _____ of the Moon.

• Moon and the Earth • around the Earth • one side • axis • the same place